

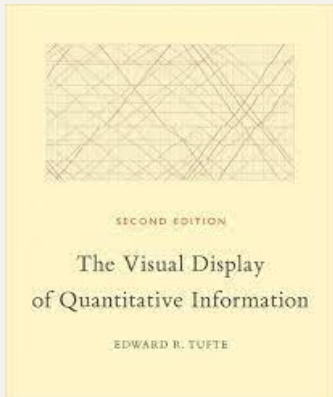
Data Visualization and Storytelling

Professor Laura Ruiz
MSc Business Analytics



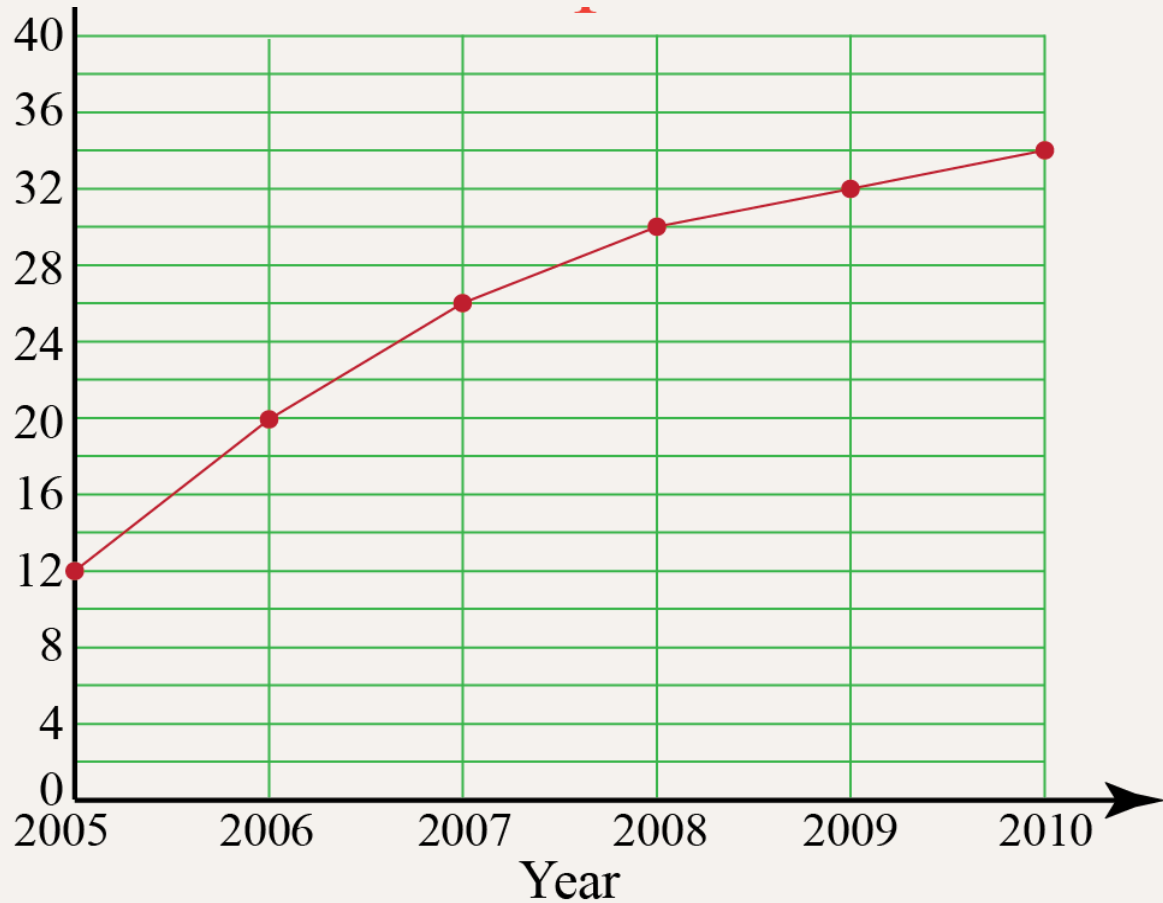
March 20, 2026

SESSION 2: The science of graphical perception & introduction to Power BI



Saket, B., Srinivasan, A., Ragan, E. D., & Endert, A. (2017). Evaluating interactive graphical encodings for data visualization. *IEEE transactions on visualization and computer graphics*, 24(3), 1316-1330.

Quick recap: Group activity

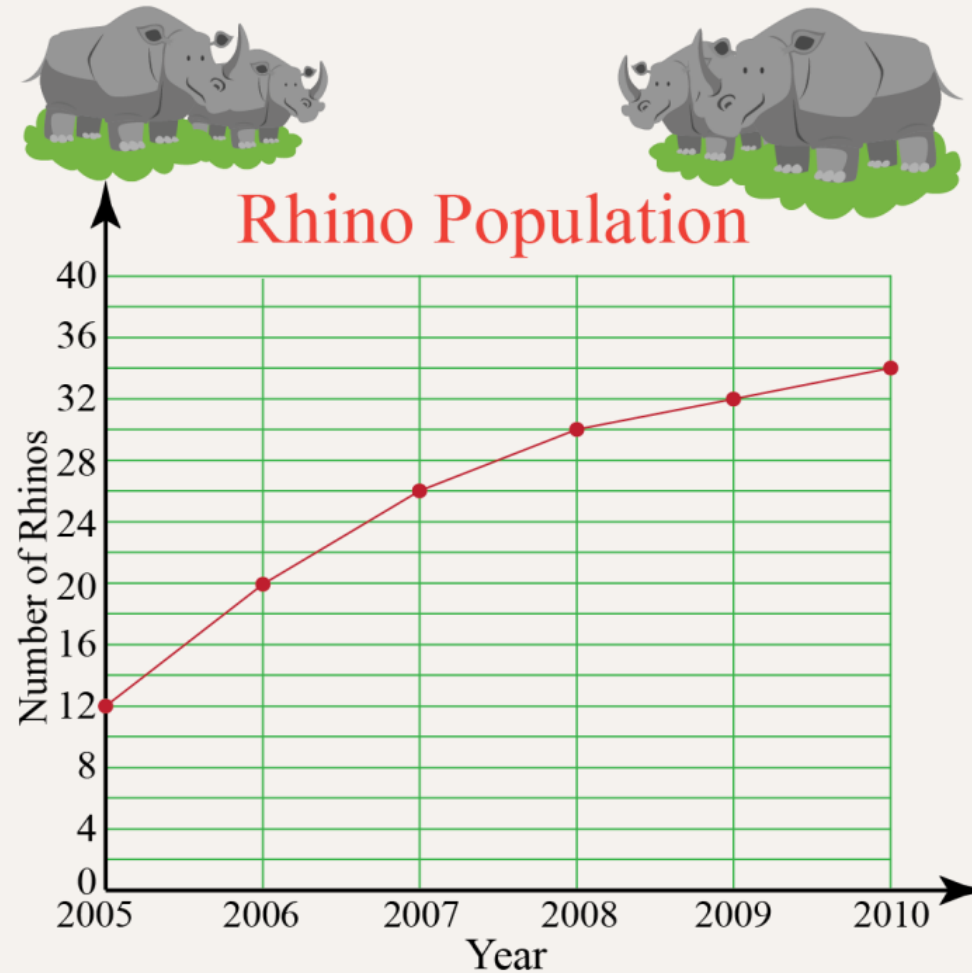


Get into small groups.

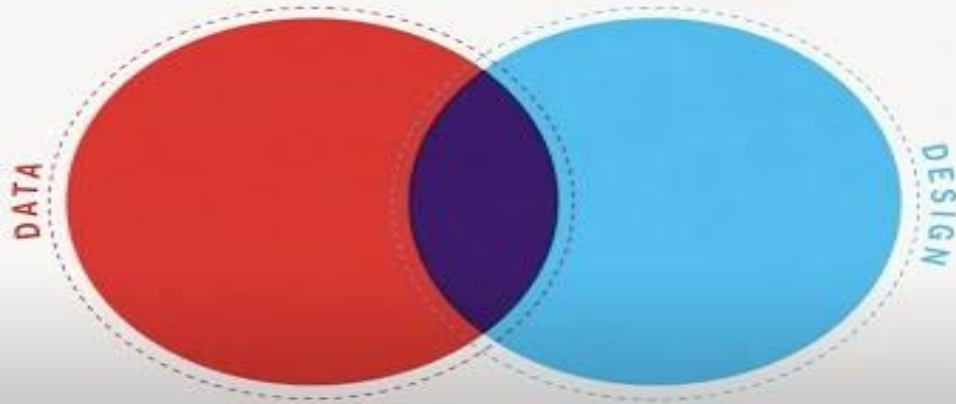
Your job: explain what's happening using only this chart.

5 minutes — then share with the class!

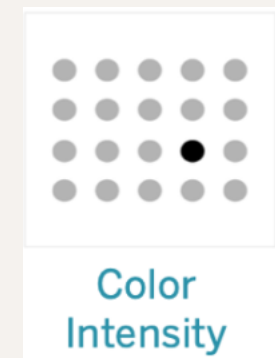
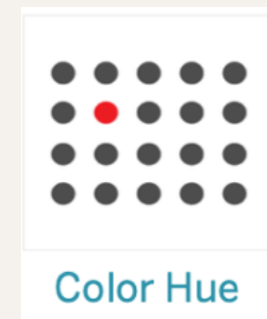
Quick recap: Group activity



Context shapes interpretation. Without context, data is just noise.



- Visual analytics leverages **pre-attentive attributes** to guide you down the most useful paths.
- Pre-attentive attributes are information we can process visually almost immediately, before sending the information to the attention processing parts of our brain.
- They are visual clues



56789

56789 color

56789 size

56789 orientation

56789 texture

The **Communication** Process: Encoding and Decoding

- In communication, the **encoding process** involves converting a message into a form that can be transmitted, such as a series of letters, numbers, shapes or symbols.
- The **decoding process** is the process of interpreting or understanding a message that has been encoded or transmitted in a particular format or language.
- The **decoding process** involves taking that encoded message and interpreting its meaning.
- **Data visualization** helps to simplify and communicate information to users, allowing them to quickly identify patterns, trends, and insights that might otherwise be difficult to discern.

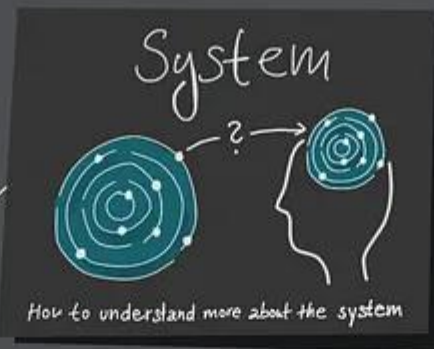
Encoders



1. 
Identify measurable
objects

2.
Define data
structure,
collect data
& apply statistics

3.
Generate
visualizations
& data products



6.
Update mental map
and story

5.
Interpret and
find meaning

4.
Crack translation
key 
Identify objects
& properties

Decoders



The Communication Process: Encoding and Decoding

“No matter how clever the choice of information and no matter how technologically impressive the encoding, a visualization fails if the decoding fails.”

William Cleveland, The Elements of Graphing Data

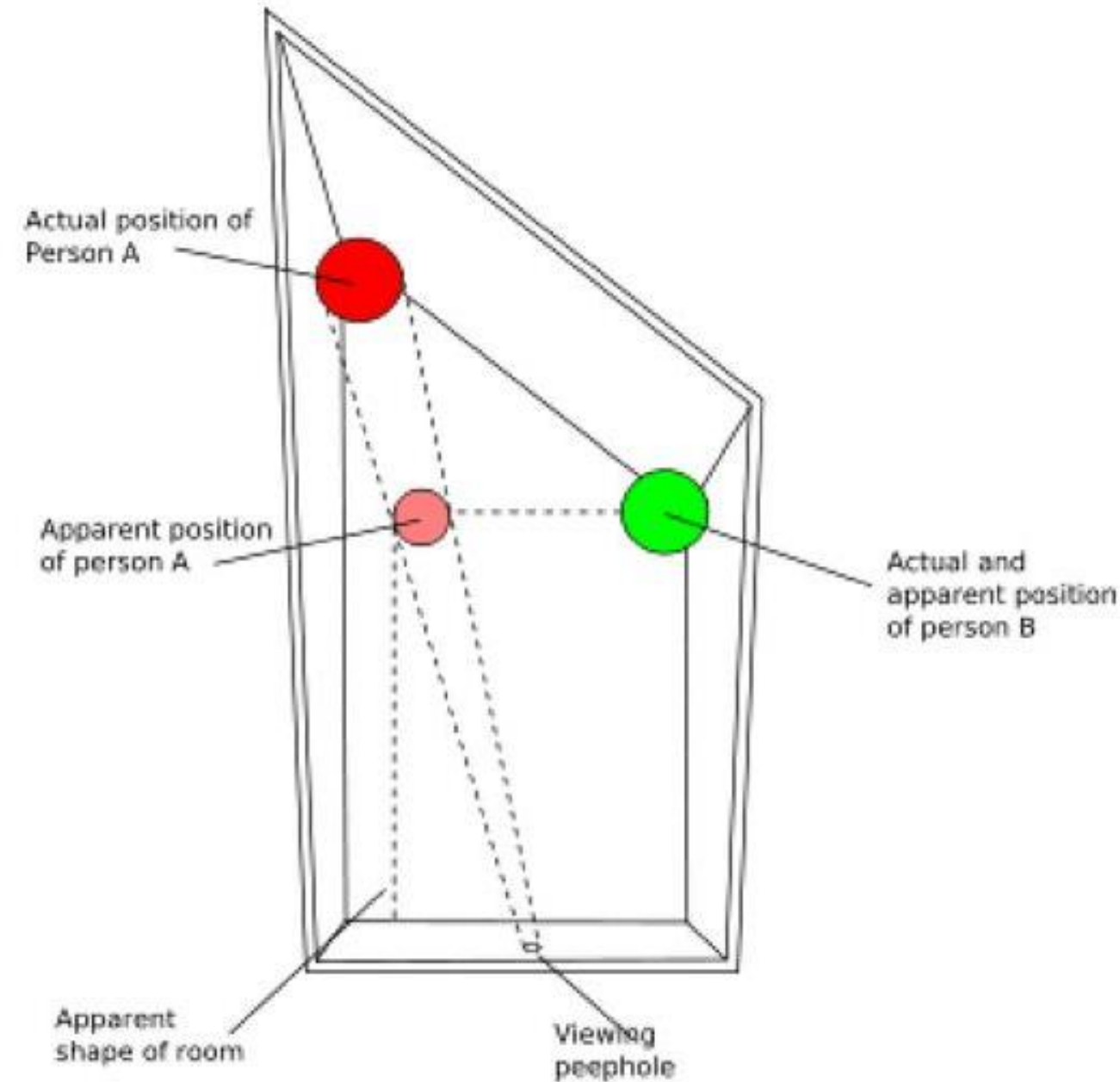






- This is called **Ames room** illusion: distorted room used to create an optical illusion of relative sizes.
- This is taken to indicate the significant role past experience has on our interpretation of our perceived world.
- Ames and subsequent researchers used this phenomenon to demonstrate the importance **of experience in perception**. We favor the mistaken perception of a normal room and wrongly see the people as different sizes.





If the **encoding is misleading** (e.g., distorted axes, inappropriate scales, 3D effects), then the viewer's **decoding** will be flawed — they will **misinterpret** the data.

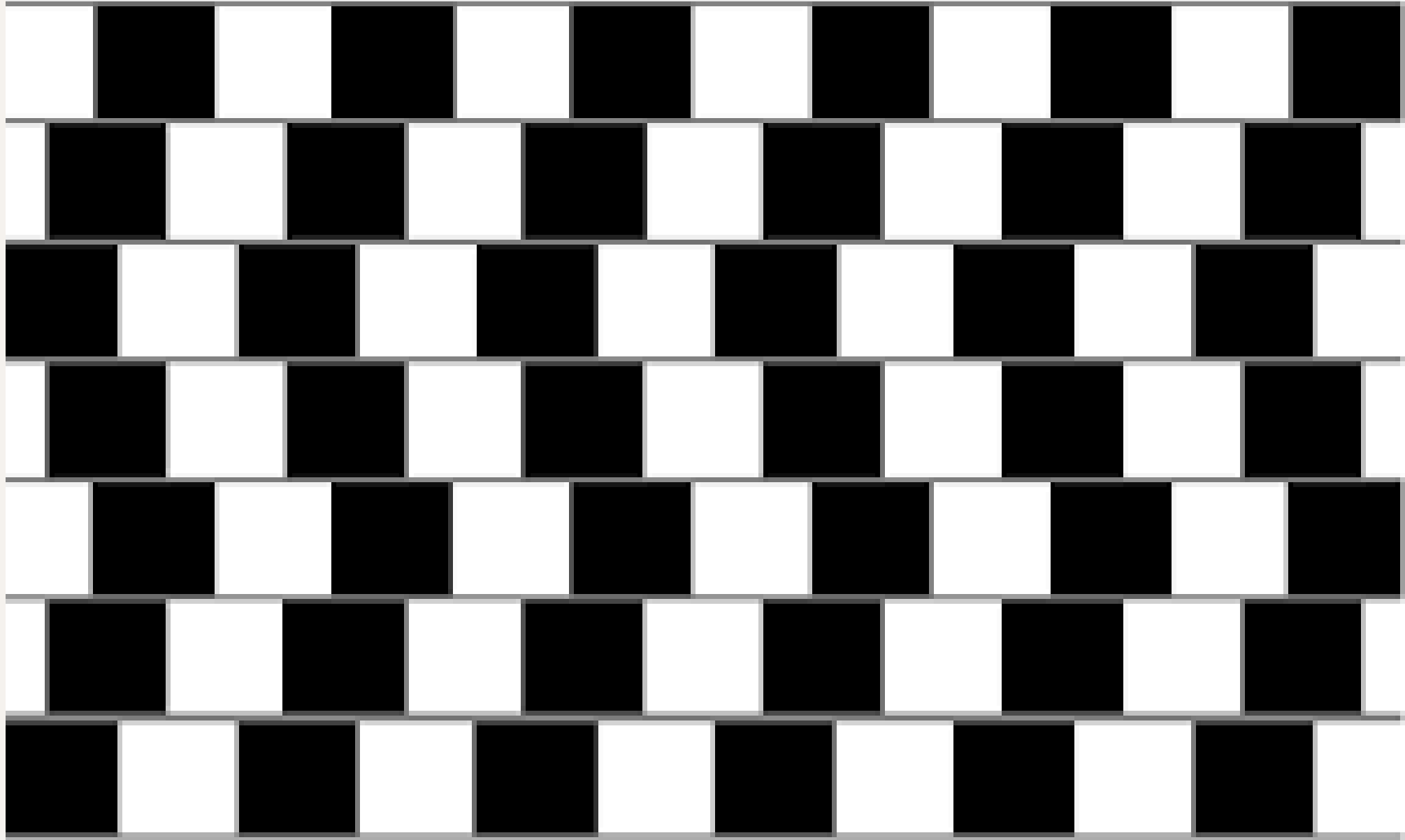
Data visualizations must be honest and clear in how they encode data, because users **rely on visual perception** to decode meaning.

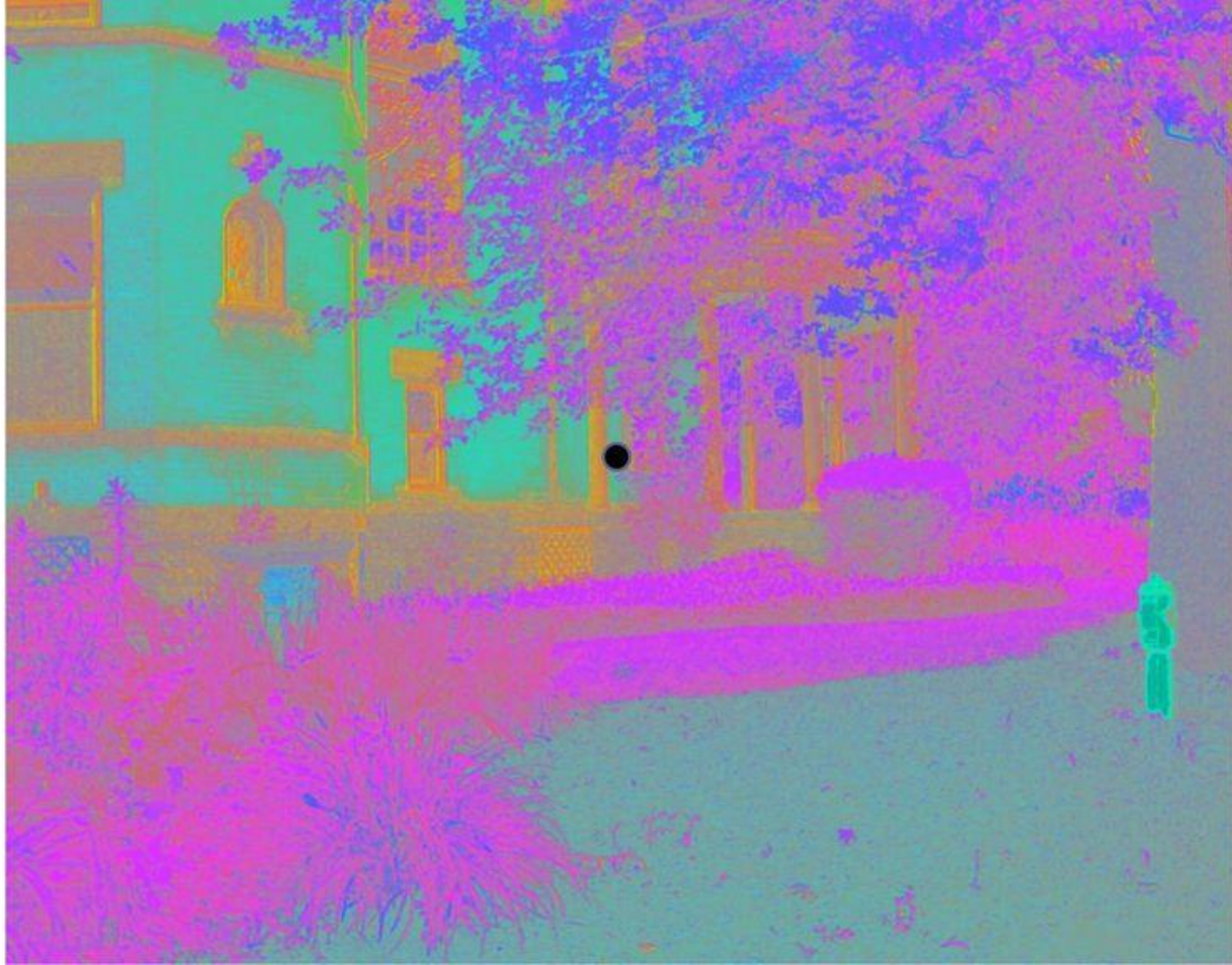
Misleading design = misleading insights.



Source: Wiki Commons (Lotus, Illinois Railroad Tracks)

The cafe wall illusion: **perception is not always reality**









How to ensure that your decoding is successful? Well...

Start with the audience

- Try to understand how the audience decode the message
- This is where we talk about **visual decoding principles**
- **Cleveland and McGill** studied what people are able to decode most accurately and ranked them in a list

Graphical perception

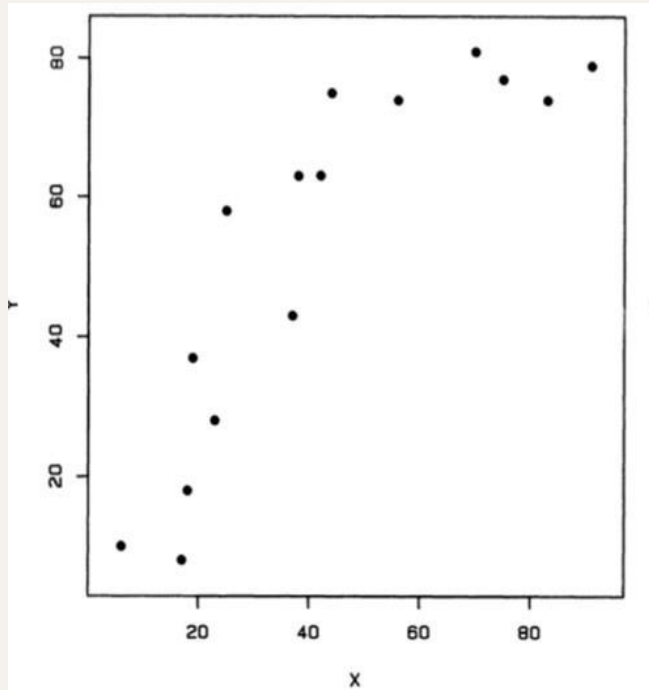
- In the context of extracting information from graphs we will call it **graphical perception**
- Graphical perception is the visual **decoding** of information encoded on graphs
- Graphs that convey their information at this unconscious level allow us to extract the information without any conscious effort on our part

What are the **perceptual tasks**?

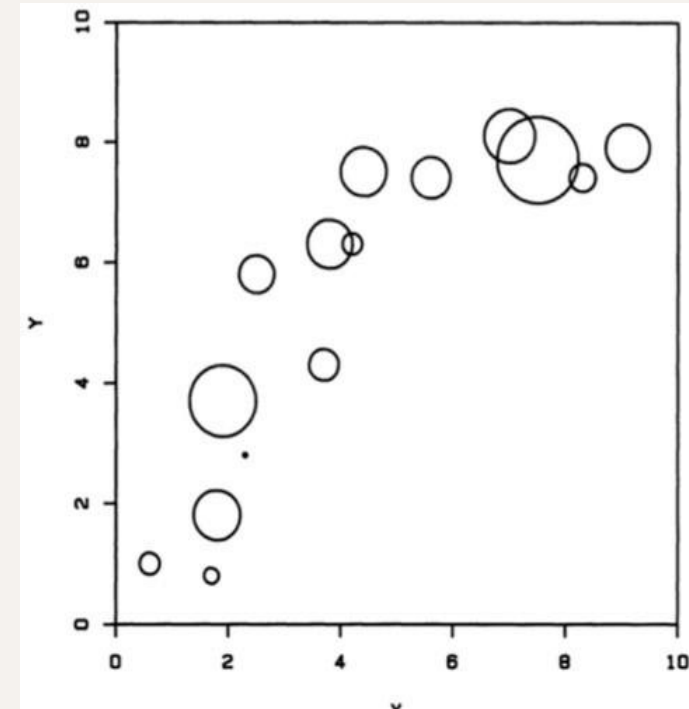
- When a user is trying to decode the information in the graph, **one of the perceptual elementary tasks has to be performed**
- **Elementary perceptual tasks will be carried out when people extract quantitative information from graphs**
- Some of these tasks are easier to perform than others
- Some of these tasks yield more accurate perceptions of the numerical quantities depicted
- Let's look at an example

Extracting quantitative information

Where you can extract more quantitative information?

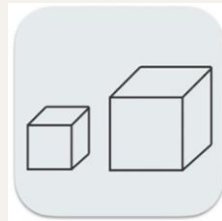
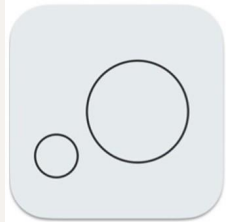
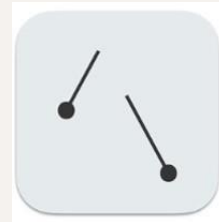
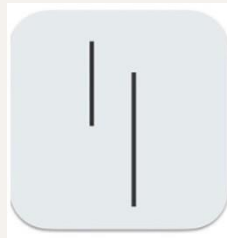
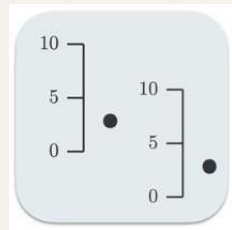
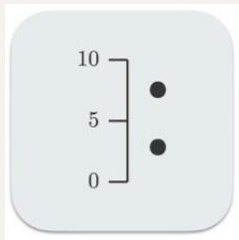


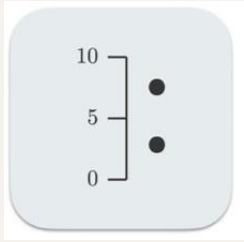
Position



Position + Area

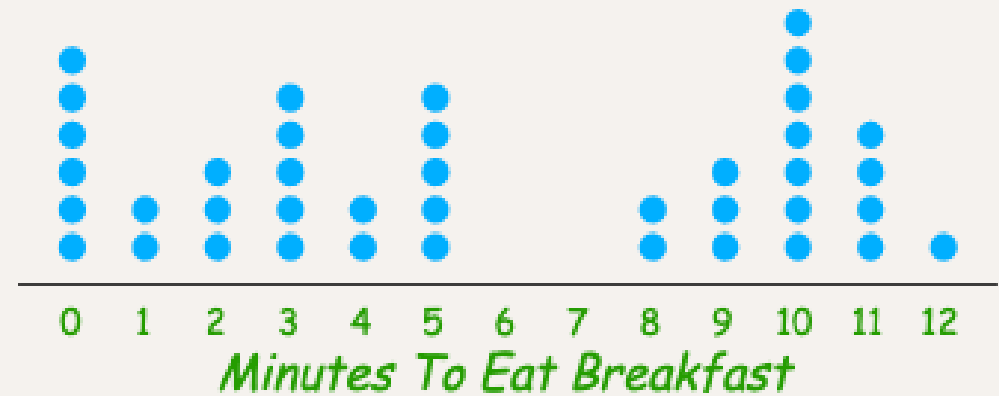
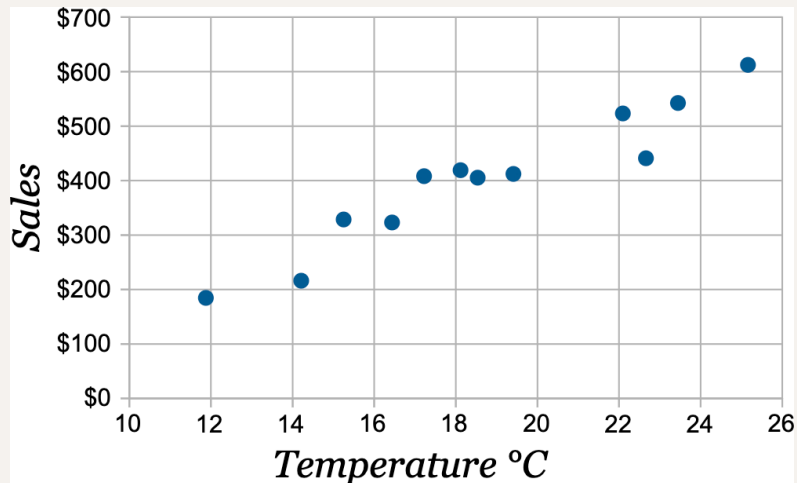
The 10 Elementary Perceptual Task our engages in

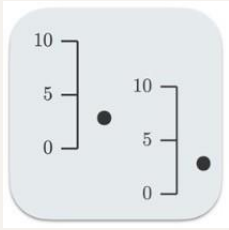




The position in a **common** scale

It involves using spatial location along a common baseline to represent data.



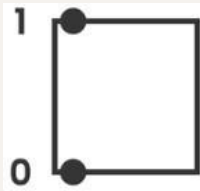


The position in **non-aligned** scales

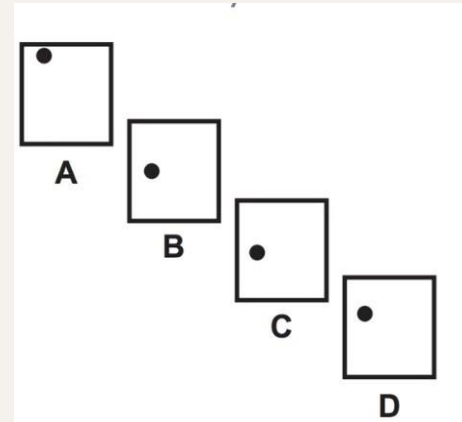
The position is still used, but the baselines differ.

This is typical in a grouped bar chart where each group has a different baseline.

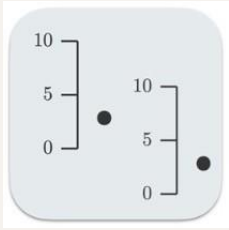
Assume the following:



Task : Let's rank the data from high to low



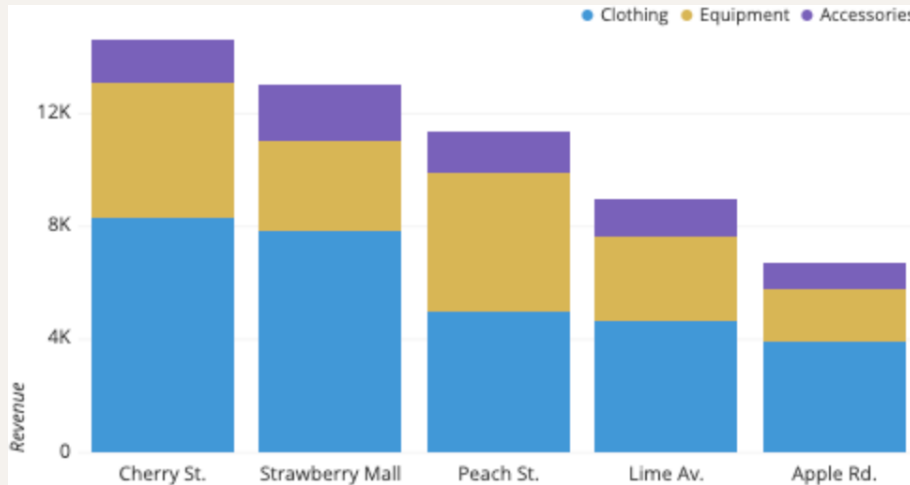
What did you notice about this task?



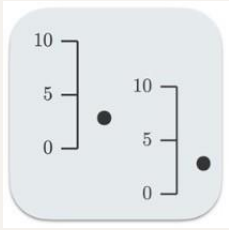
The position in **non-aligned** scales

The position is still used, but the baselines differ.

This is typical in a grouped bar chart where each group has a different baseline.



We want to move to a **stacked bar chart** when we care about the relative decomposition of each primary bar based on the levels of a **second categorical** variable.

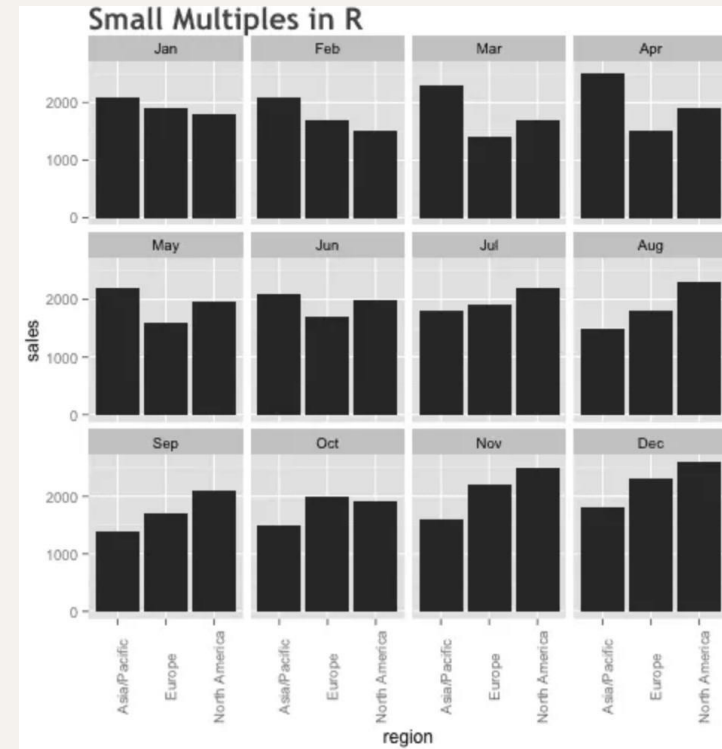


The position in **non-aligned** scales

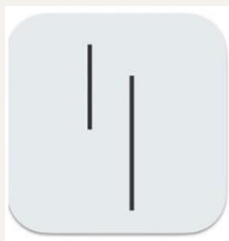
The position is still used, but the baselines differ.

This is typical in a grouped bar chart where each group has a different baseline.

Small multiples are effective because they take a cluttered chart and break it into pieces. Instead of confusing the reader with multiple pieces of data crammed into a single chart, small multiples break out the data, allowing the reader to see pieces of it individually, and compare them to each other.

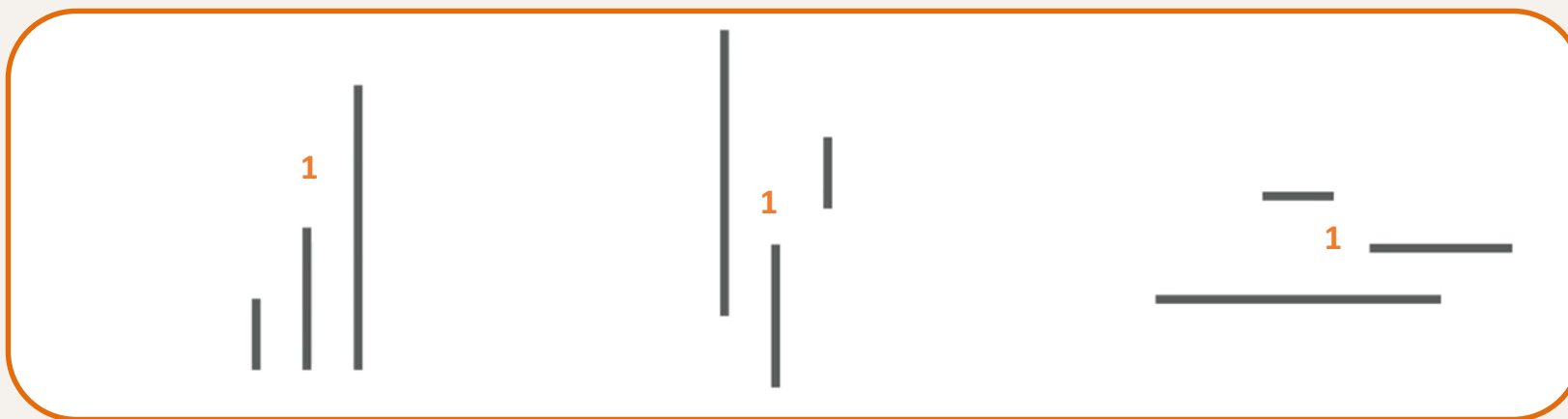


Plot Examples



Length

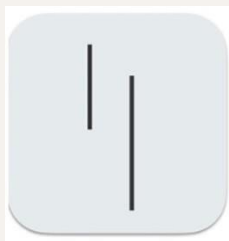
Task: Can you estimate the length of each of the lines?



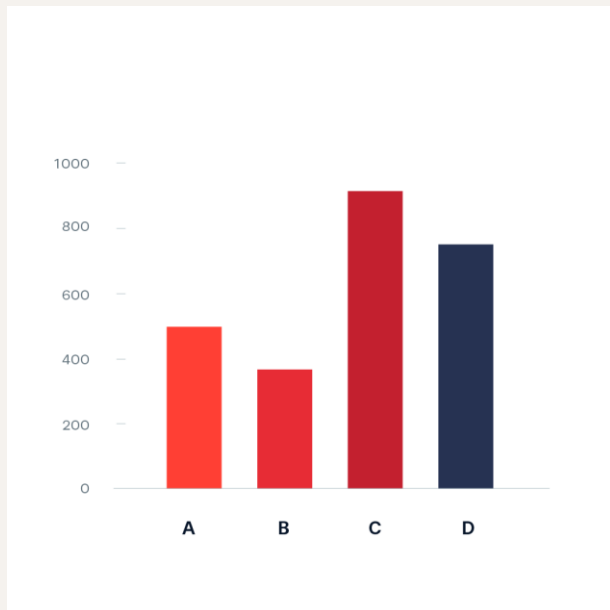
Common Scale
+ Length

Non-Aligned Scale
(vertical)
+ Length

Non- Aligned Scale
(horizontal)
+ Length



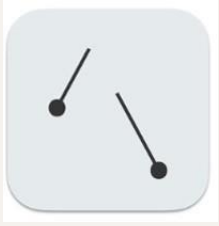
Length



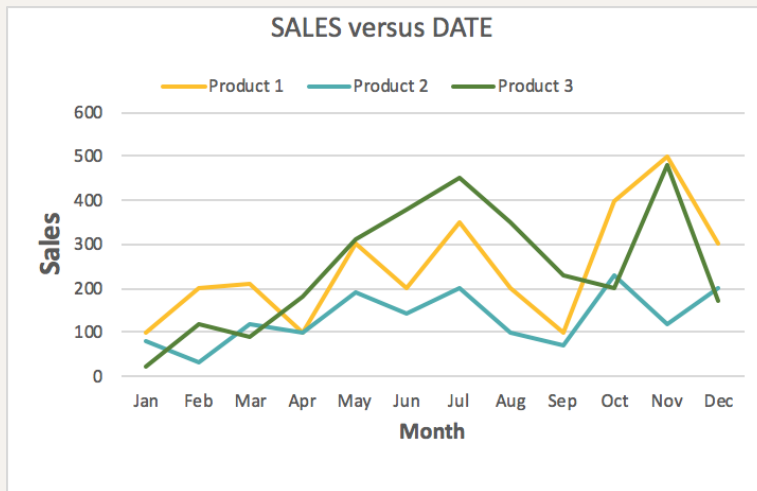
The standard way to compare the size of things. Must always start at 0 on the axis.

Best Practice note:

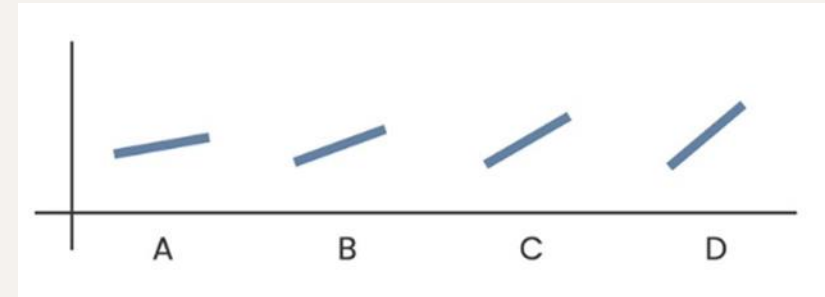
- Keep bars thin
- Keep bars consistent



Direction

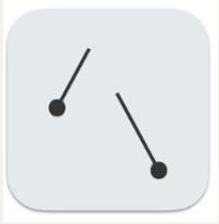


Length+ Direction

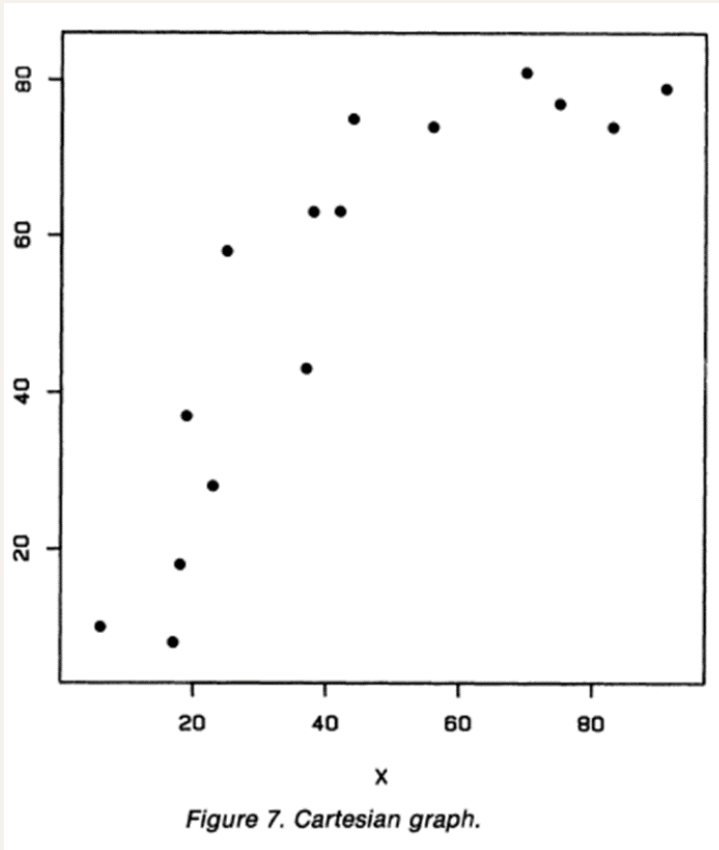


Common Scale + Direction

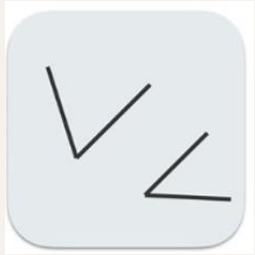
Plot Examples



Direction

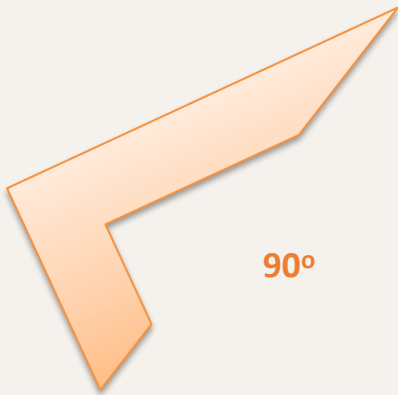


- Cartesian graph (scatterplot) of paired values of two variables, x and y .
- The values of x can be visually extracted by perceiving position along a scale, in this case the horizontal axis.
- The y values can be perceived in a similar manner.
- The real power of a cartesian graph is not on one's ability to perceive the x and y values separately but rather to **understand the relationship of x and y .**



Angle

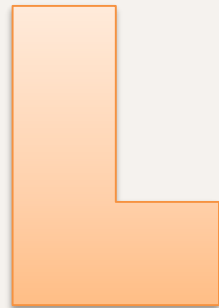
Task: Can you estimate the value of these angles?



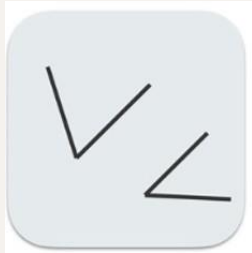
90°



90°



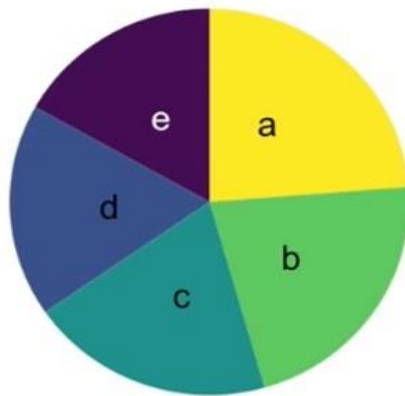
90°



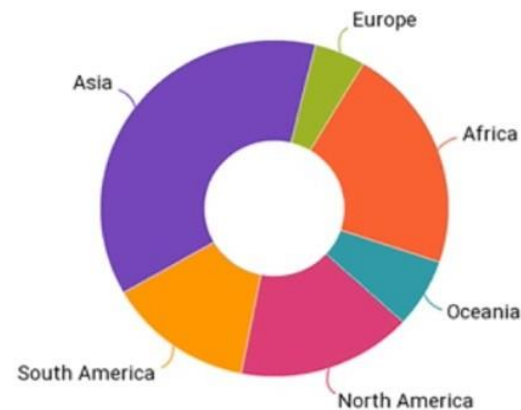
Angle

Used to represent **proportionality** (use them with caution). The problem is called the **Jastrow illusion** (an optical illusion that makes us perceive the outer of two visible rings as being smaller than the inner ring). We can **misread** double donut charts.

Pie Chart

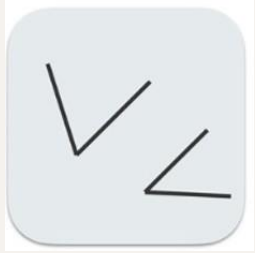


Doughnut Chart



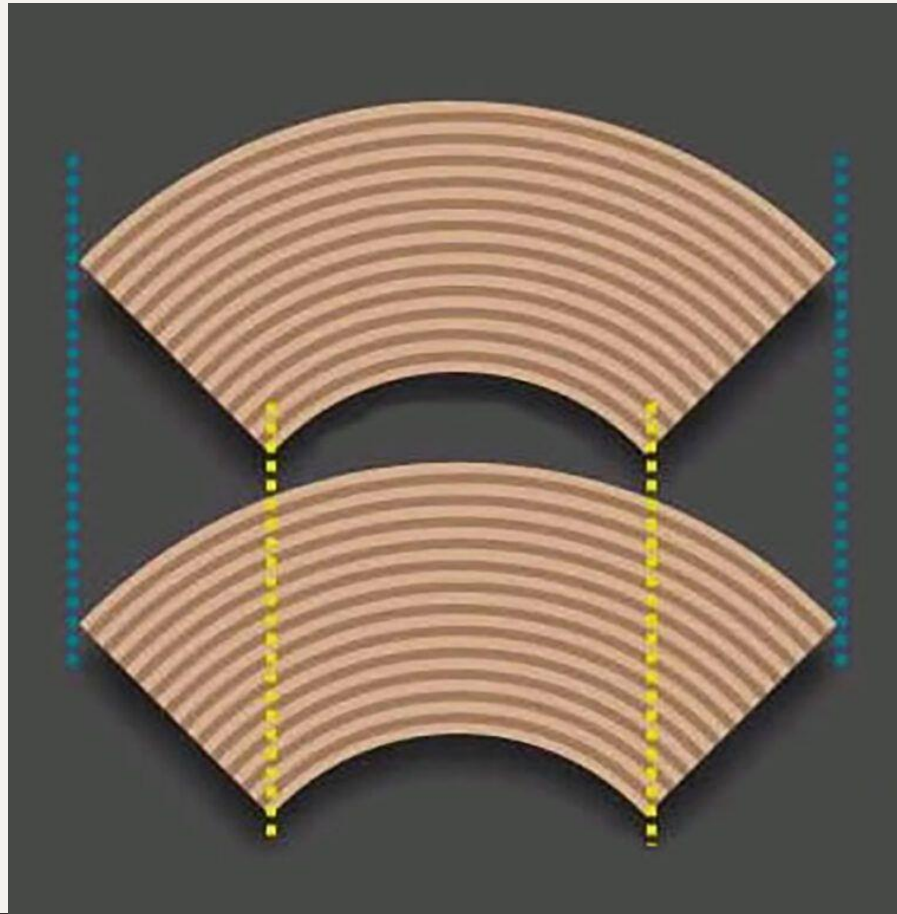
Sunburst Chart

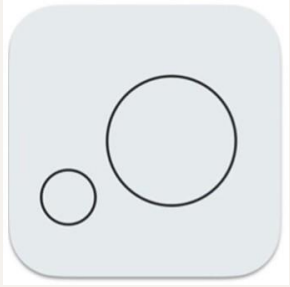




Angle

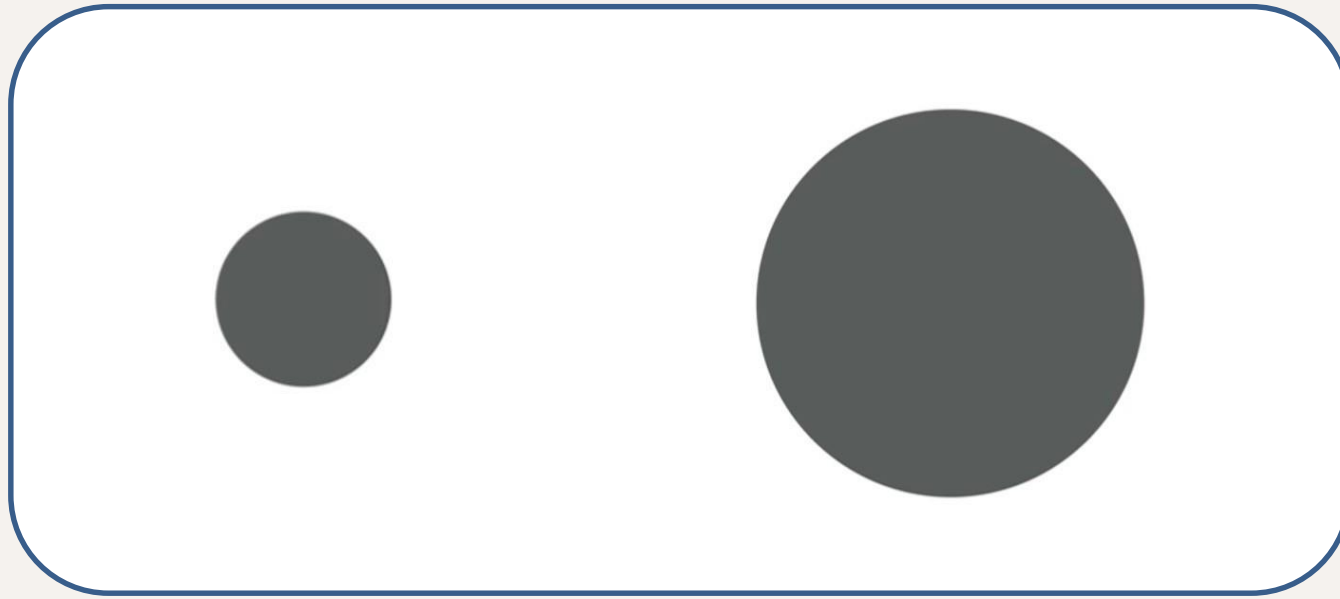
Used to represent **proportionality** (use them with caution). The problem is called the **Jastrow illusion** (an optical illusion that makes us perceive the outer of two visible rings as being smaller than the inner ring). We can **misread** double donut charts.





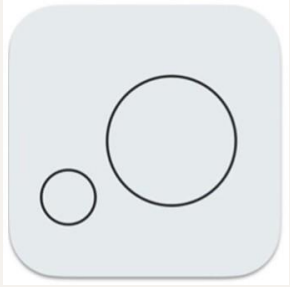
Area

Task: Can you estimate how much B is bigger than A?

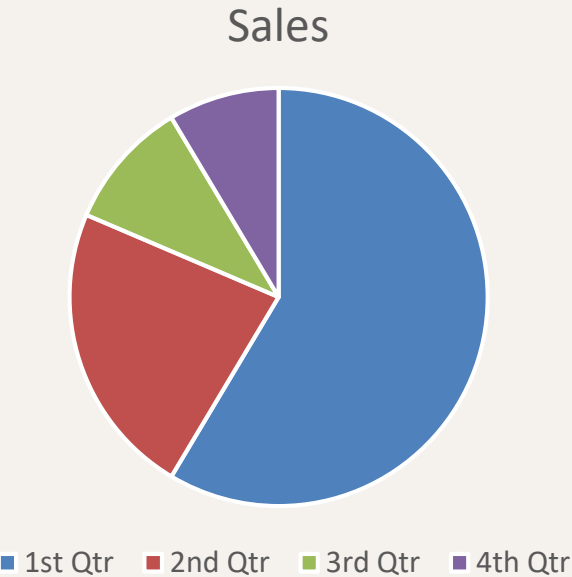


Area = 1

Area = 6



Area

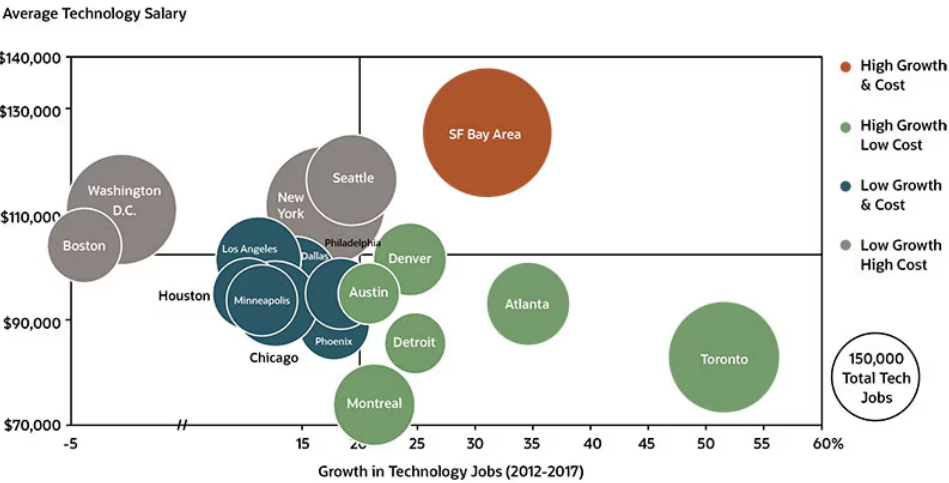


Pie Chart

Bubble Chart

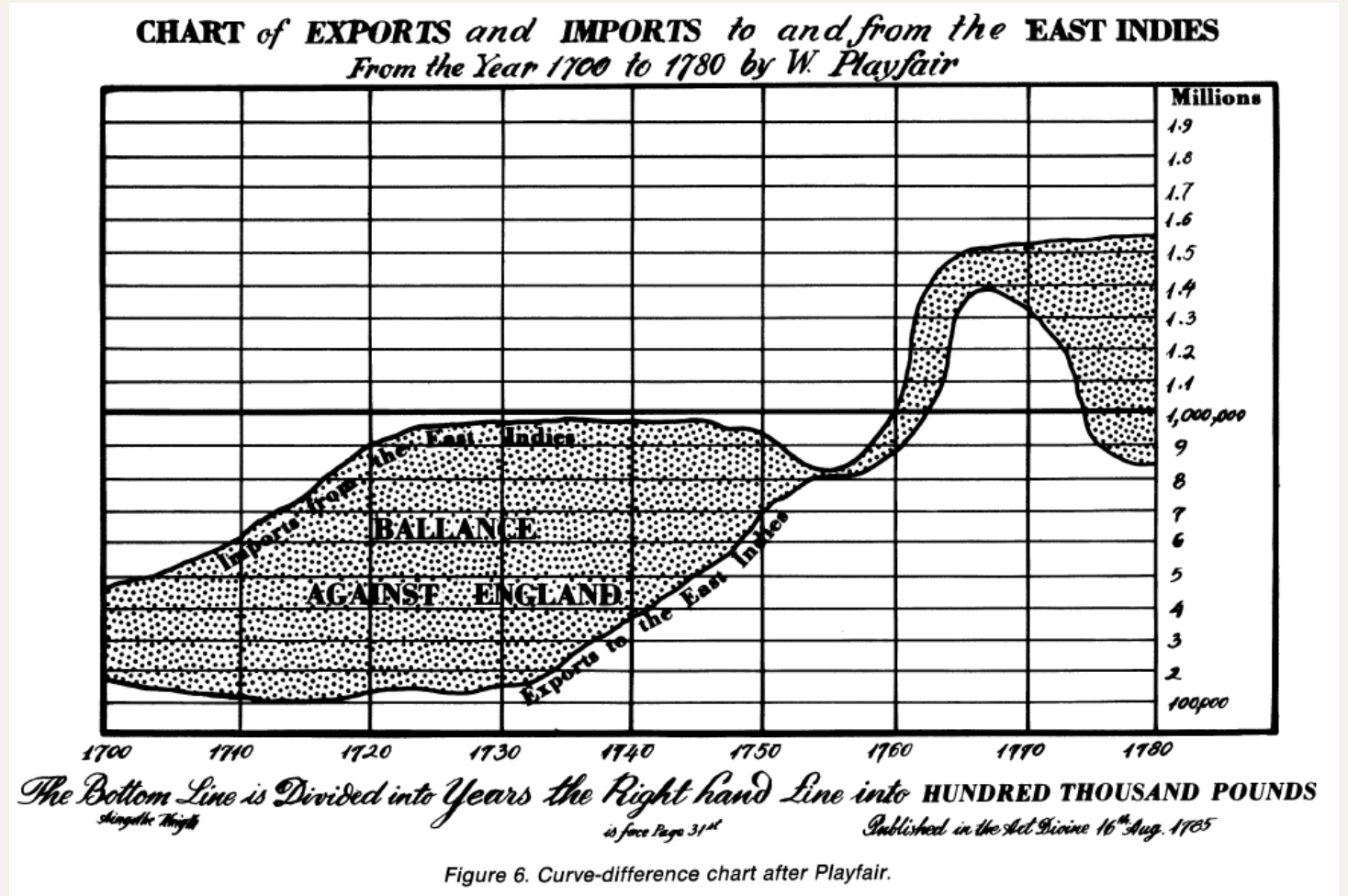
Technology Markets In North America

While San Francisco is still the largest market for technology jobs, Toronto is the fastest growing and it offers companies significantly lower wage costs.





Curve

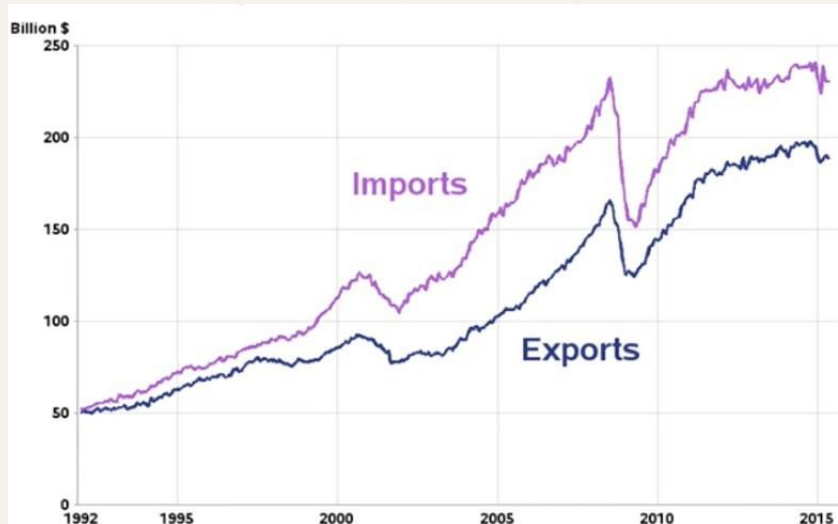


Plot Examples



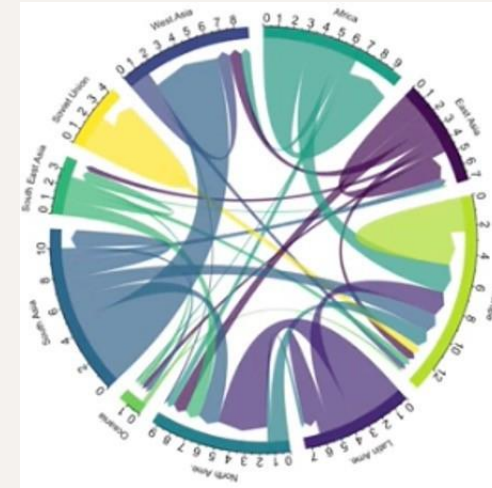
Curve

Line plot

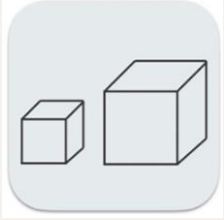


The standard way to show a changing time series. If data are irregular, consider markers to represent data points

Chord plot

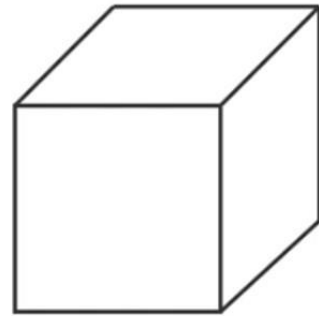


A chord plot represents flows or connections between entities (called nodes). Each entity is represented by a fragment on the outer part of the circular layout. The size of the arc is proportional to the importance of the flow

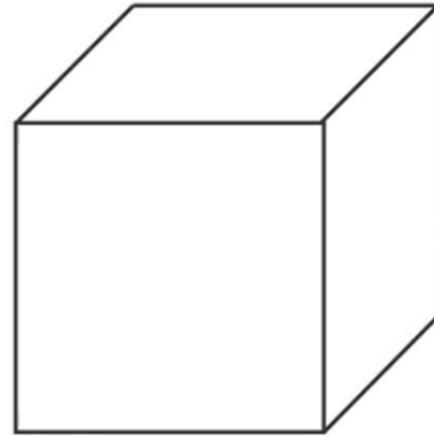


Volume

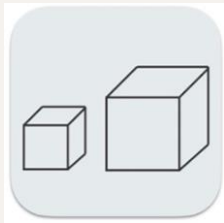
Task: How much bigger is B compared to A?



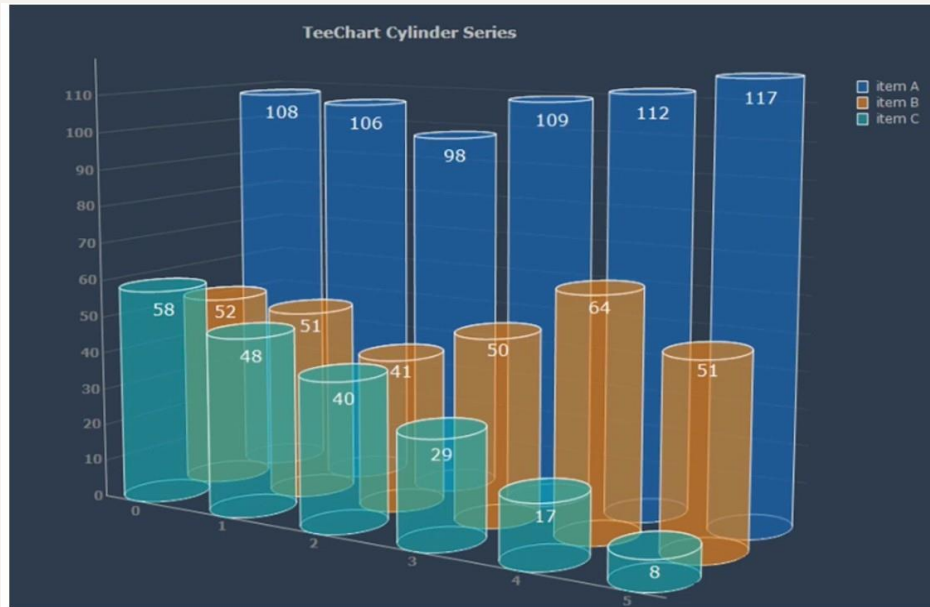
A = 1



B = ?
B = 3.37



Volume



Our brains are not good at estimating or comparing **3D shapes**, especially **cylinders or blocks** in perspective.

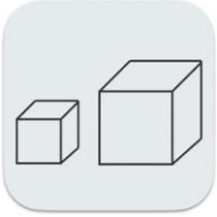
Small differences in height may be **exaggerated or flattened** due to the 3D effect.

Volume can be tricky

What 5 Ounces of Wine Really Looks Like

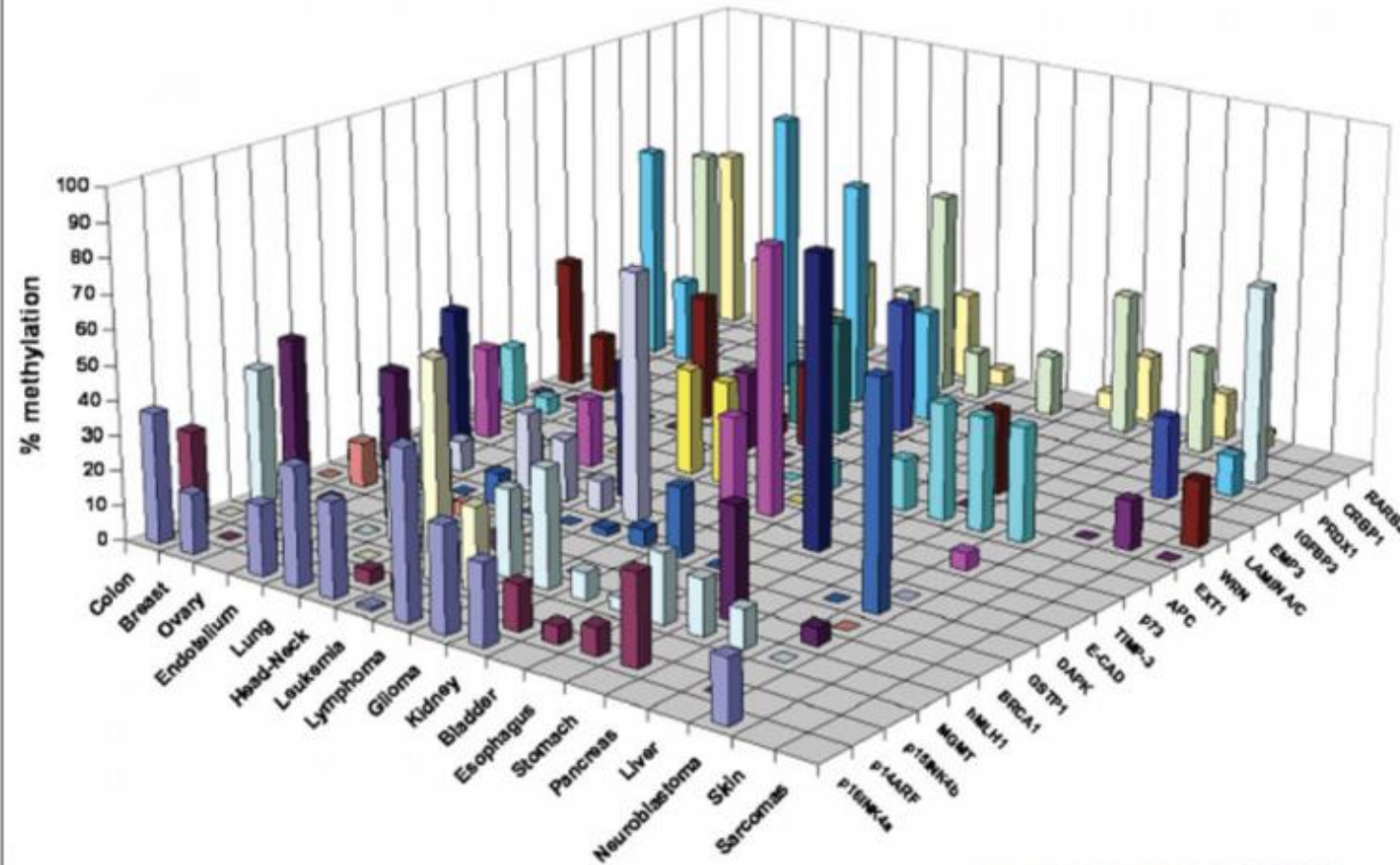
When health experts say a glass of wine is good for your health, they're talking about a 5-ounce serving. Depending on the size and shape of the glass you choose, that amount can look very different.





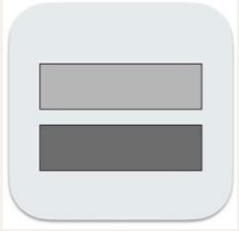
Volume

A CpG Island Hypermethylation Profile of Human Cancer



Hum. Mol. Genet. (2007) 16:R50-59

Plot Examples



Shading

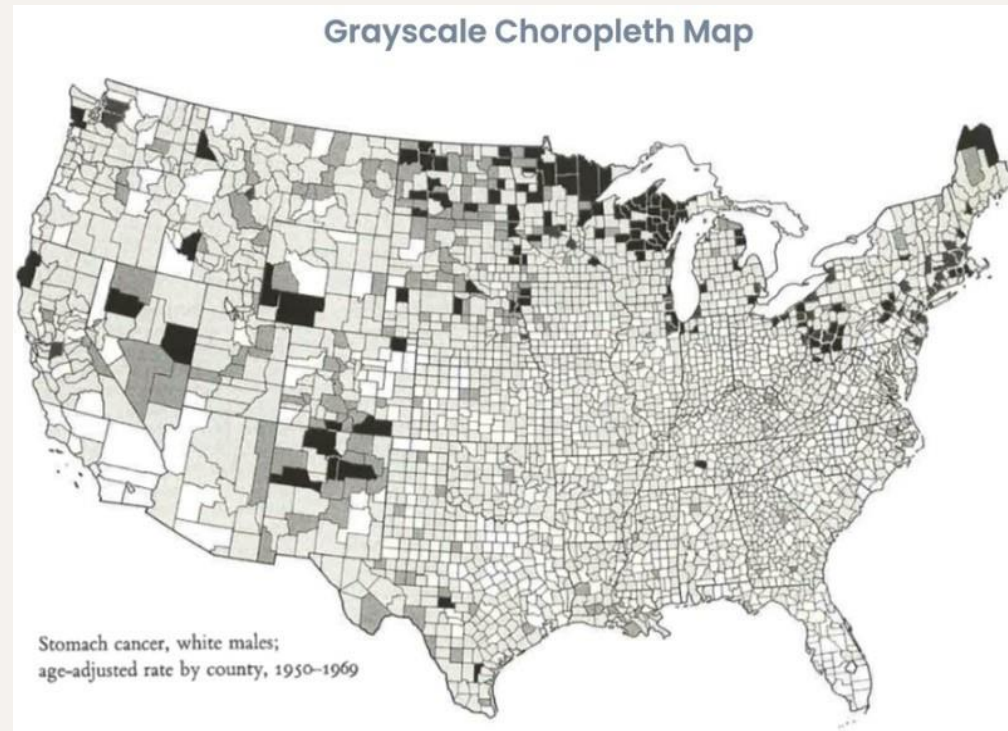
Task: Rank the example below from lighter to darker



We are not good at judging shading.
A color might look darker or lighter depending on what's
around it → Simultaneous contrast



Shading

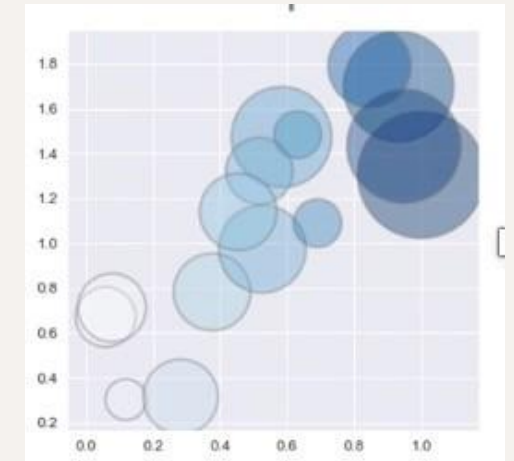
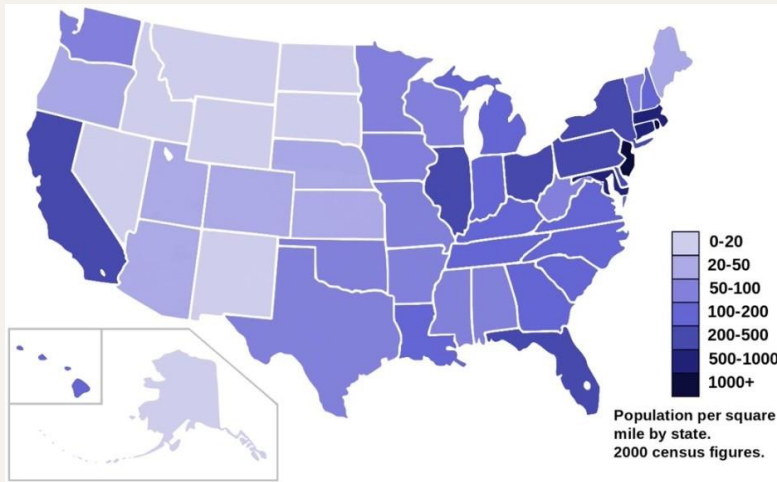




Color saturation

Decode the color → match it to a legend → extract the value

Darker States indicate higher population



Plot Examples

THE USE OF COLOR IN DATA VISUALIZATION

SEQUENTIAL

color is ordered from low to high



DIVERGING

two sequential colors with a neutral midpoint



CATEGORICAL

contrasting colors for individual comparison



HIGHLIGHT

color used to highlight something



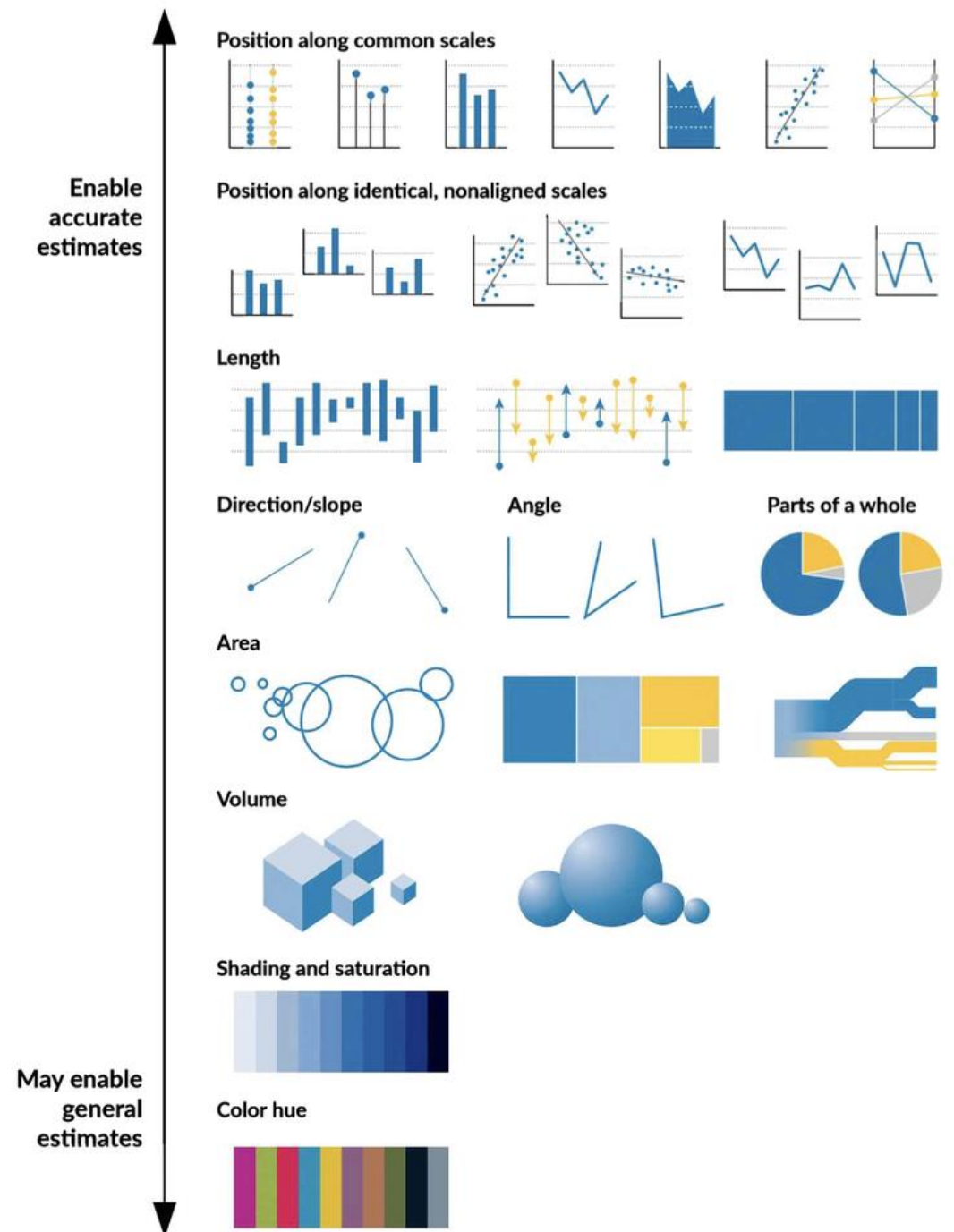
ALERT

color used to get reader's attention



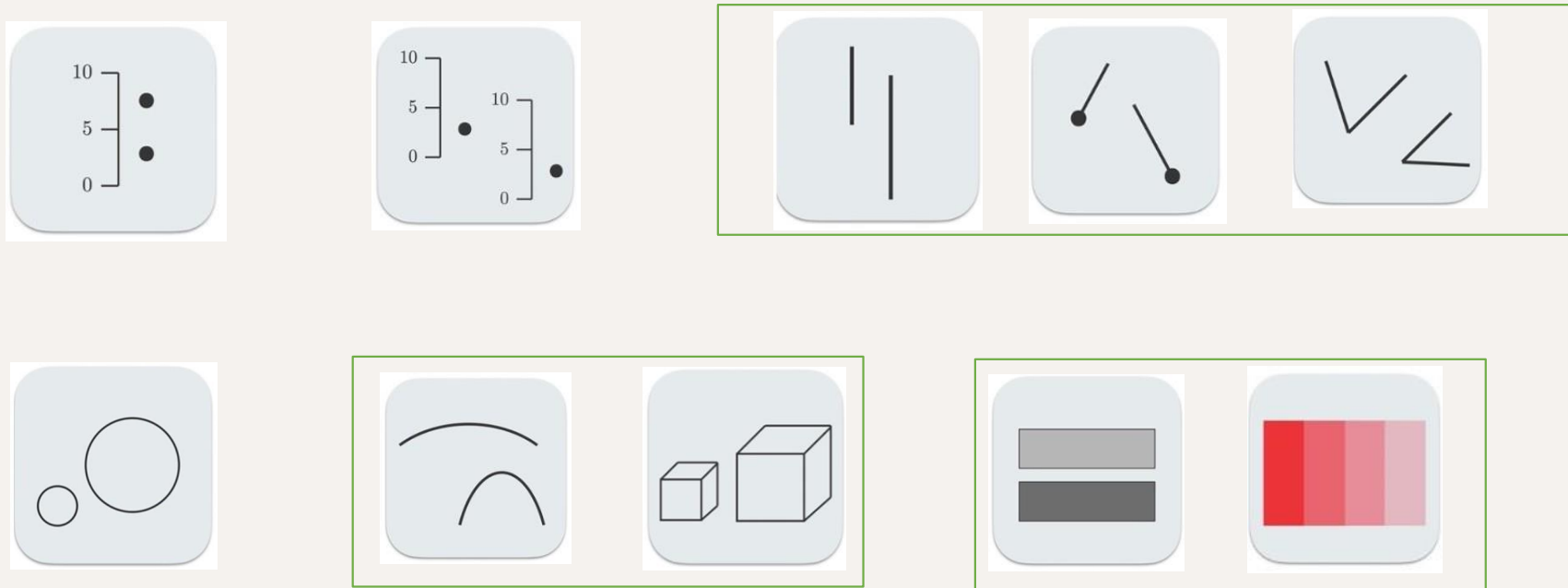
Can we order these perceptual tasks based on how accurately people can extract the quantitative information?

(Cleveland and McGill's goal)



Perceptual Ranking Diagram. This image is based on Alberto Cairo (2016) from research by Cleveland and McGill (1984), Heer, Bostock, and Ogievetsky (2010), and others.

Ranked from best to worst



One elementary perceptual task is taken to be more accurate than another if it leads to human judgments that come closer to the actual encode quantities.

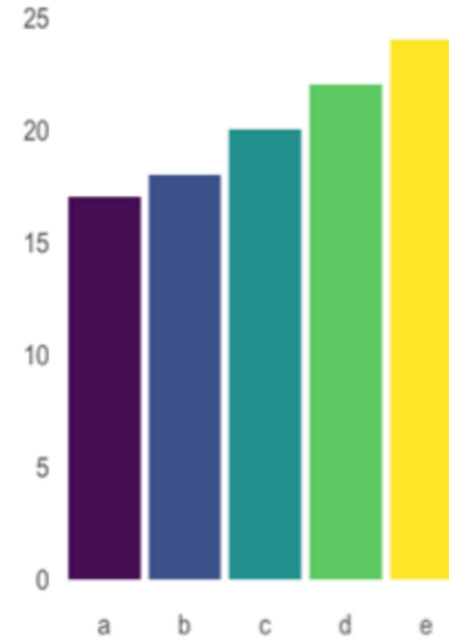
Let's test your perceptual tasks

Which of these is more complex to read?

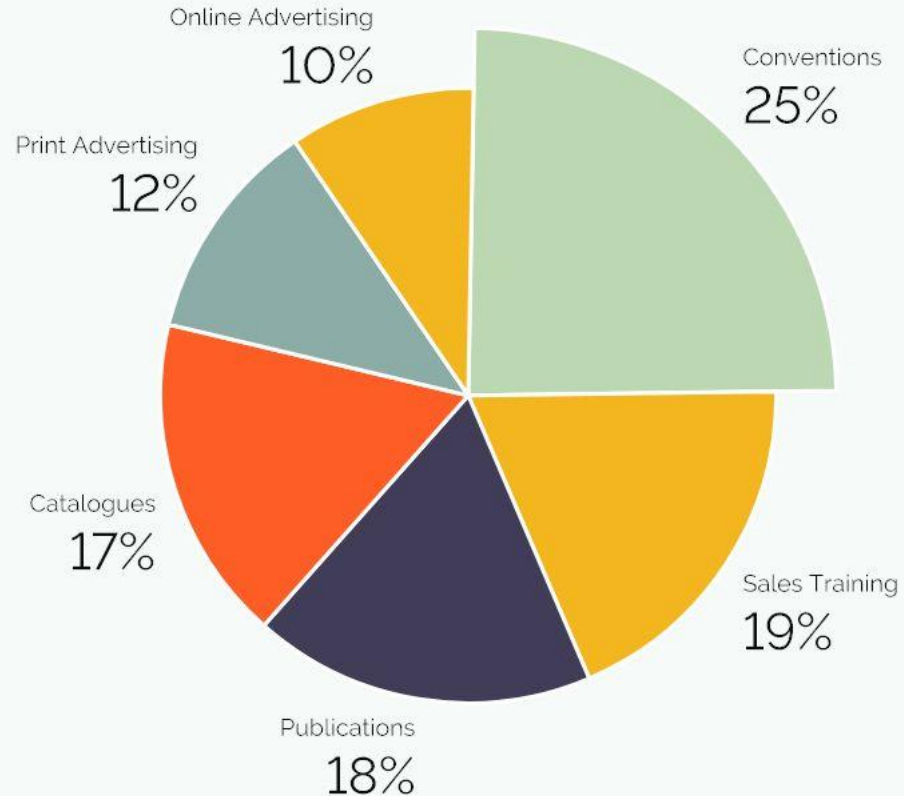
A



B

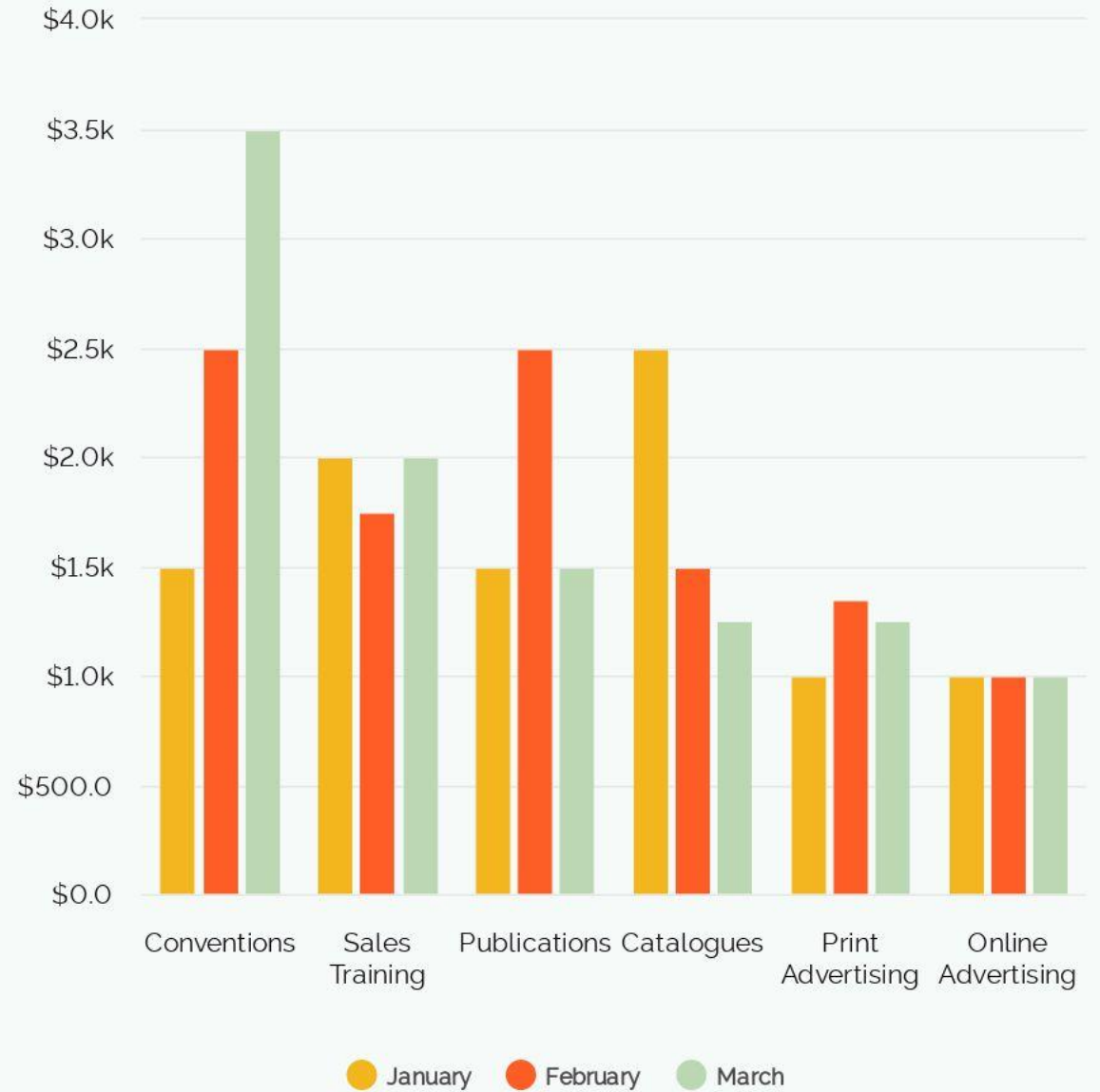


Top Marketing Expenditures



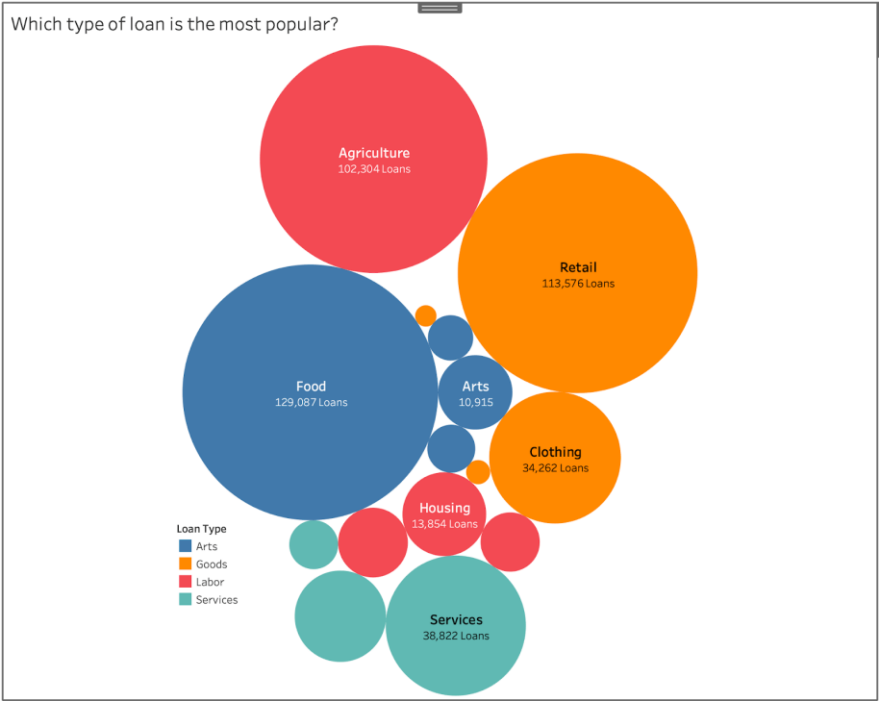
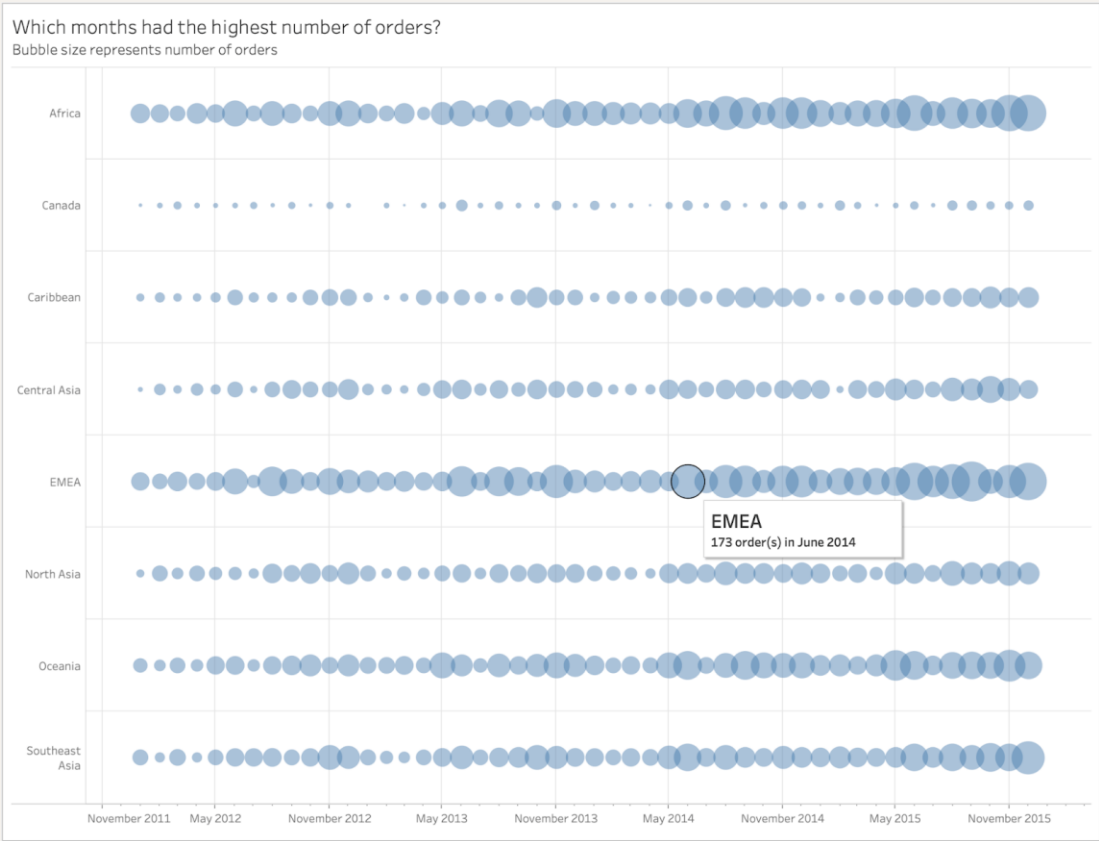
How would you plot this info + these categories in January, February and March?

Expenditures by Month



Let's test your knowledge

What perceptual tasks do we need to decode these plots?



Let's test your knowledge

What perceptual tasks do we need to decode these plots?



- a) Color and curve
- b) Color and angle
- c) Color and common scale

Let's test your knowledge

What perceptual tasks do we need to decode these plots?



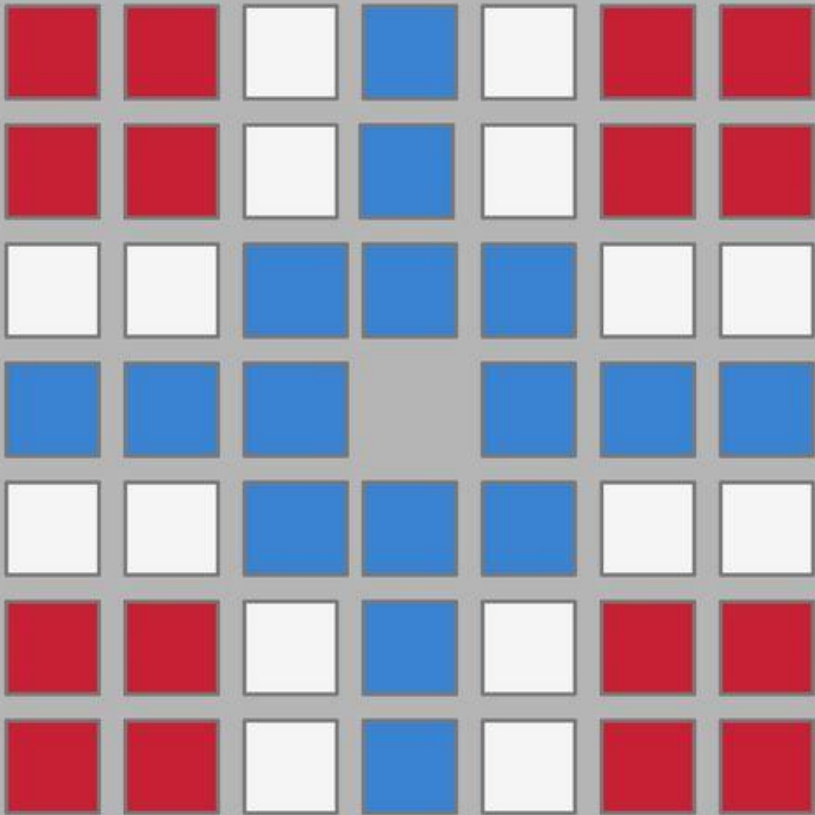
a) Color and curve

b) Color and angle

c) Color and common scale

Let's test your knowledge

Are there more blue, white
or red squares?



The Chart Doctor

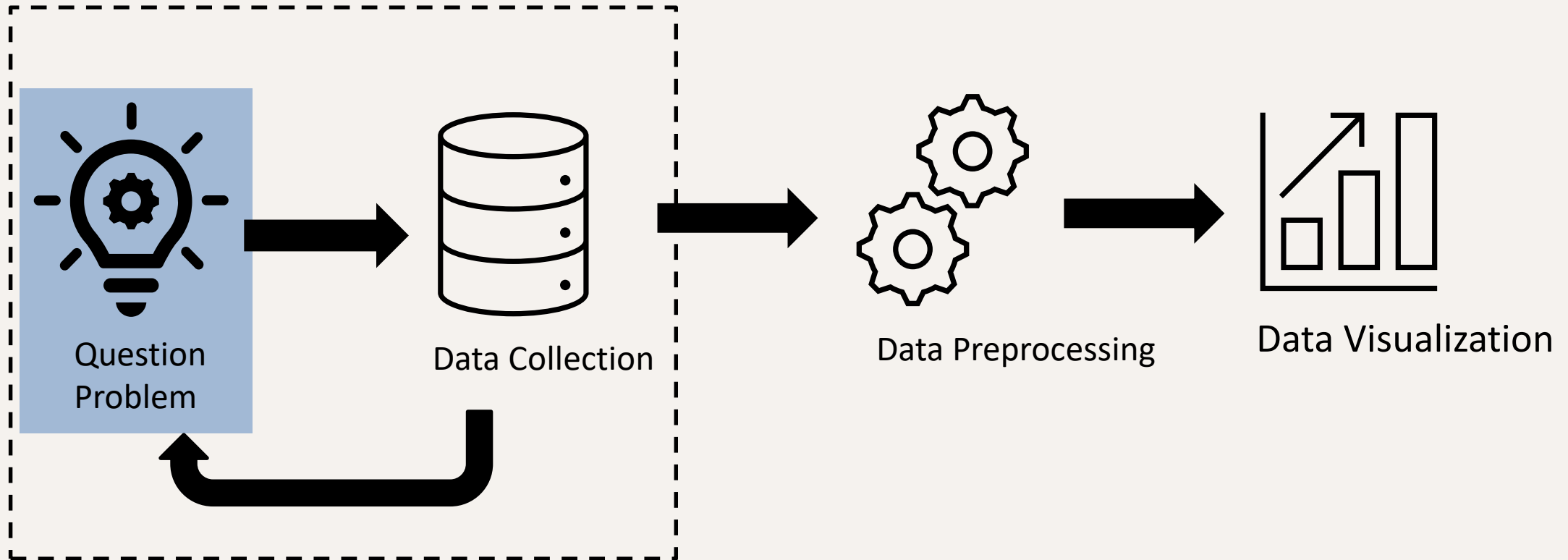
The science behind good charts

Take part in our interactive experiment to boost your chart-making confidence



<https://ig.ft.com/science-of-charts/>

Data Visualization Lifecycle



Data collection and data sources

Data is our mechanism of communication

Where is our data coming from?

Primary (surveys, sensors) vs. Secondary (databases, APIs, public datasets)

What data do we have available?

Volume, variety, velocity — what's usable vs. what needs transformation?

Is our data trustworthy?

Provenance, recency, completeness — can we stake a decision on it?

Types and sources of data



Structured

- Organized in rows & columns
- Predefined schema (data types, field names)
- Examples: Excel tables, SQL databases, CSV files
- Easy to query, filter, and aggregate
- → **Power BI's sweet spot**



Semi-Structured

- Has some structure, but flexible/nested
- Examples: JSON, XML, web APIs
- Requires Power Query to flatten/expand
- Common in cloud & REST API sources
- → **Needs transformation first**



Unstructured

- No predefined model — free text, images, video
- Examples: PDFs, social media posts, emails
- Not directly usable in Power BI
- Requires external NLP/ML preprocessing
- → **Rare in standard BI workflows**

Types and sources of data

Excel (.xlsx / .xls)

Most common source in business. Supports tables, named ranges, and multiple sheets. Power BI imports each sheet/table separately.

💡 Tip: Always format data as an Excel Table (Ctrl+T) before importing — it makes column headers and new rows auto-detected.

CSV / Text (.csv, .txt)

Simple flat files with delimiter-separated values. No formatting, no formulas — just raw data. Universally compatible.

💡 Tip: Watch out for encoding issues (UTF-8 vs. UTF-16) and inconsistent delimiters when importing from different systems.

JSON / XML (.json, .xml)

Hierarchical/nested formats common in APIs and web exports. Require expansion steps in Power Query to flatten into tables.

💡 Tip: Use 'Expand to New Rows' or 'Extract Values' in Power Query to unnest JSON arrays into usable columns.

PDF (.pdf)

Power BI can extract tables from PDFs — but quality depends heavily on PDF structure. Works best on machine-generated (not scanned) files.

💡 Tip: If extraction looks off, try exporting the source document to Excel or CSV instead — it'll save significant cleanup time.

Types and sources of data

SQL Server / Azure SQL

Microsoft's relational DB. Most seamless Power BI integration — native connector, DirectQuery support.

MySQL / PostgreSQL

Popular open-source relational DBs. Common in web apps and startups. Require credentials + gateway for on-premise.

Oracle / IBM Db2

Enterprise-grade relational databases. Common in large corporations; Power BI connects via ODBC or native connector.

Key Concepts

Import Mode

Data copied into Power BI. Faster reports, works offline. Must refresh to update.

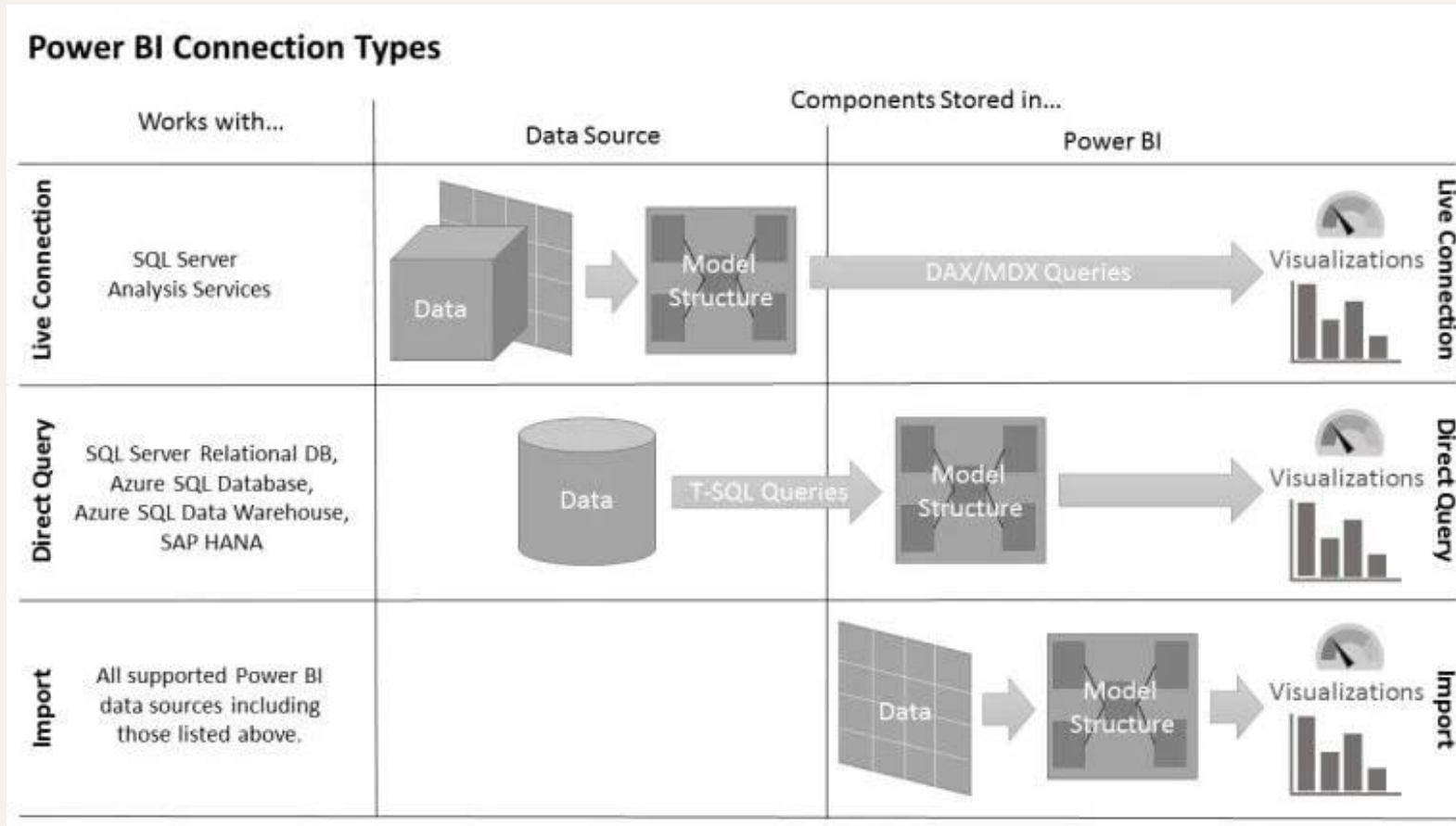
DirectQuery

Data stays in the DB. Always live, but slower — each visual sends a query.

On-Premise Gateway

Required to connect Power BI Service to local databases. Acts as a secure bridge.

Types and sources of data



A bridge that allows Power BI (cloud) to connect to local (on-premise) data sources

Power BI does NOT store the data.
Every time you click or filter → Power BI queries the database live

The Role of Data Preprocessing



Remove

Delete rows/columns with too many missing values or irrelevant fields



Correct

Fix typos, standardize formats, resolve inconsistencies



Structure

Split, merge, and reshape columns to match your analysis needs



Filter

Select relevant subsets; remove duplicates and outliers



Transform

Convert units, aggregate by period, normalize values

Cleaning Data – Power BI tools



Power Query Editor

The main cleaning environment in Power BI Desktop. Access via Home → Transform Data. Every transformation is recorded as a step — non-destructive.



Data View

Inspect your tables after transformations. Check data types, verify values, and spot remaining issues before building visuals.



Column Quality & Profile

In Power Query: View → Column Quality shows % valid/error/empty. Column Profile shows distribution — great for spotting outliers.

A reminder on field types



Discrete Data

- Result of a counting process — whole numbers / integers
- Example: "How many Toyotas did we sell last year?"
- Creates headers in the view when placed on rows/columns
- Finite, individually separate and distinct values



Continuous Data

- Takes any value within a range — measured on a scale
- Example: Temperature, revenue, distance
- Creates an axis in the view when placed on rows/columns
- Power BI always aggregates continuous fields!

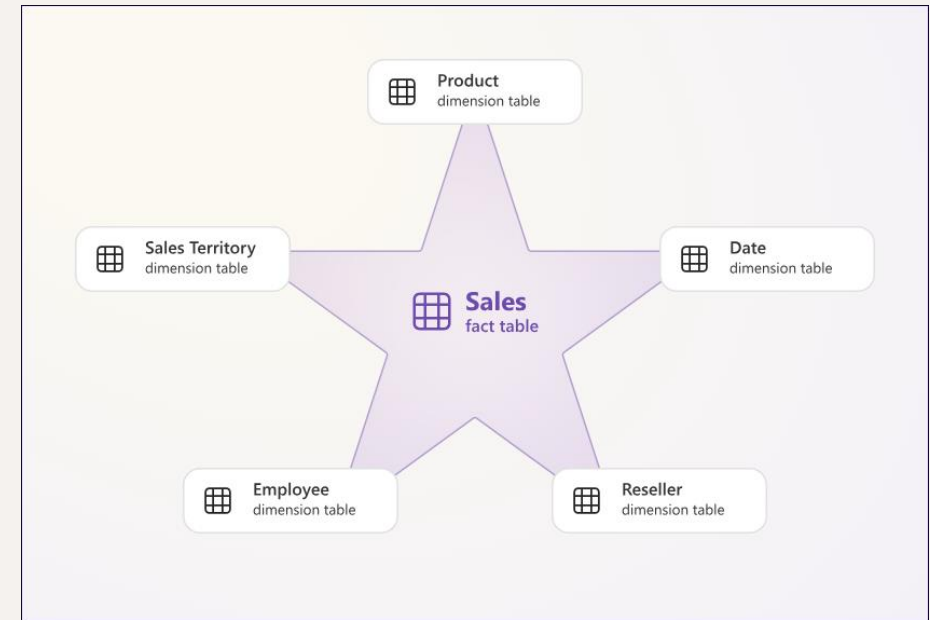
Asynchronous intro: Key concepts

Star schema overview

Star schema is a mature modeling approach widely adopted by relational data warehouses. It requires modelers to classify their model tables as either *dimension* or *fact*.

Dimension tables describe business entities—the *things* you model. Entities can include products, people, places, and concepts including time itself. The most consistent table you'll find in a star schema is a date dimension table. A dimension table contains a key column (or columns) that acts as a unique identifier, and other columns. Other columns support filtering and grouping your data.

Fact tables store observations or events, and can be sales orders, stock balances, exchange rates, temperatures, and more. A fact table contains dimension key columns that relate to dimension tables, and numeric measure columns. The dimension key columns determine the *dimensionality* of a fact table, while the dimension key values determine the *granularity* of a fact table.

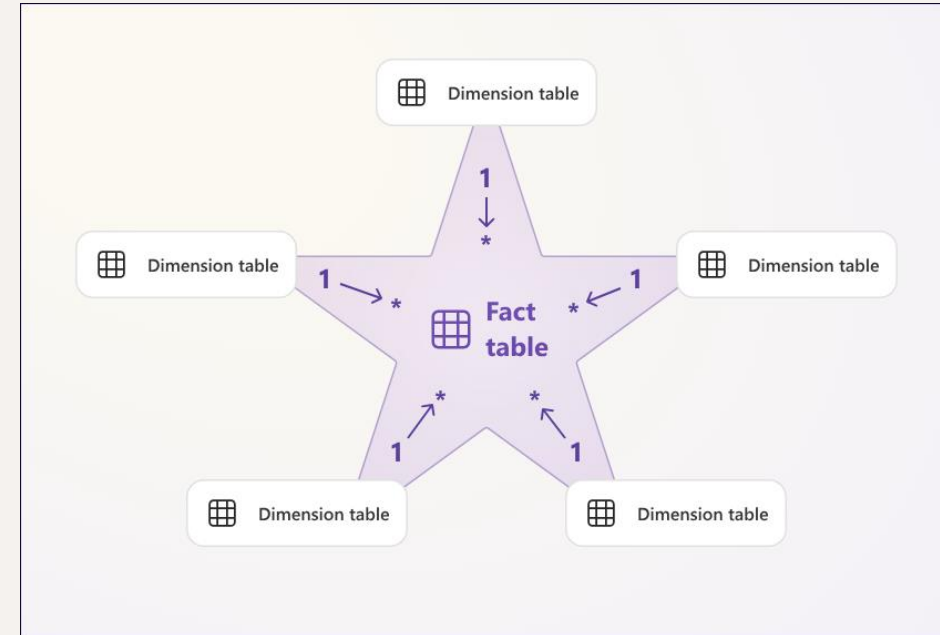


- Dimension tables enable *filtering* and *grouping*.
- Fact tables enable *summarization*.

Asynchronous intro: Key concepts

Star schema relevance to Power BI semantic models

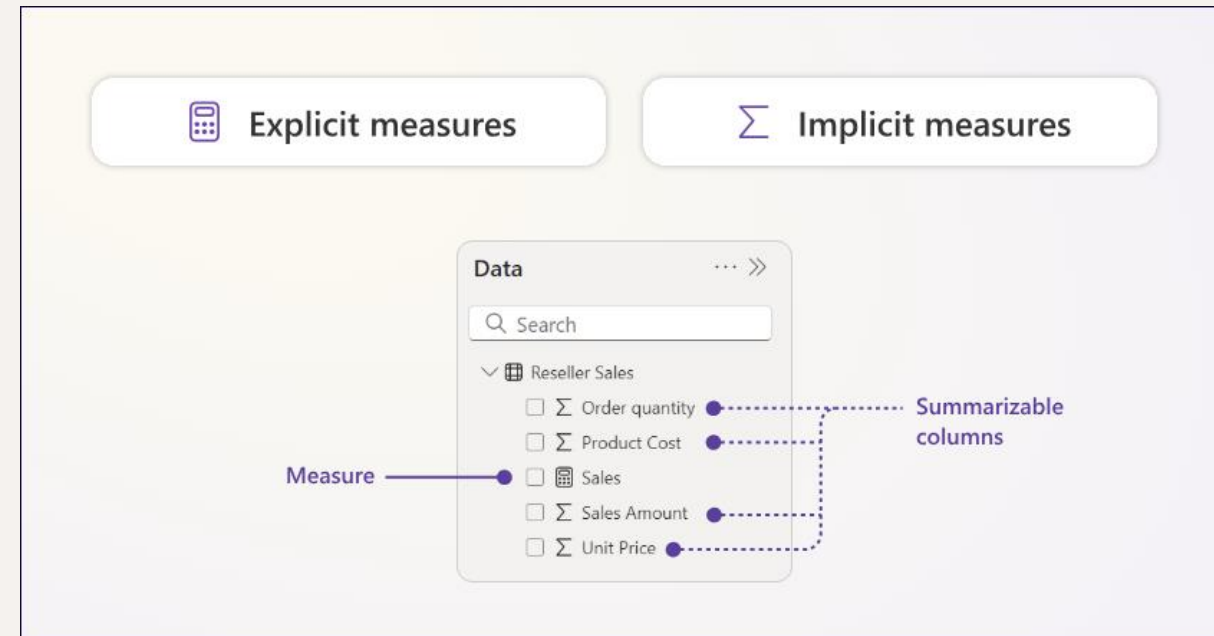
There's no table property that modelers set to set the table type as dimension or fact. It's in fact determined by the model relationships. A model relationship establishes a filter propagation path between two tables, and it's the cardinality property of the relationship that determines the table type. A common relationship cardinality is *one-to-many* or its inverse *many-to-one*. The "one" side is always a dimension table while the "many" side is always a fact table.



Asynchronous intro: Key concepts

On Measures

- **Explicit measures** are expressly created and they're based on a formula written in Data Analysis Expressions (DAX) that achieves summarization. Measure expressions often use DAX aggregation functions like SUM, MIN, MAX, AVERAGE, and others to produce a scalar value result at query time (values are never stored in the model).
- **Implicit measures** are columns that can be summarized by a report visual or Q&A. They offer a convenience for you as a model developer, as in many instances, you don't need to create (explicit) measures.



In the Data pane, explicit measures are represented by the calculator icon while implicit measures are represented by the sigma symbol (Σ).